

THE CLASSIFICATION OF THE GRUENBERG–KEGEL GRAPHS OF FINITE SOLVABLE CUT GROUPS

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ABSTRACT. A finite group is cut if the central units of its integral group ring are trivial; equivalently, for every element g , each generator of $\langle g \rangle$ is conjugate to g or to g^{-1} . Bächle, Kiefer, Maheshwary and del Río showed that a graph with four vertices which is the Gruenberg–Kegel graph of a finite solvable cut group is one of seven explicit graphs, three of which occur, and asked which of the remaining four occur. Two of them were excluded by Debón, García-Lucas and del Río. We show that the other two do not occur either. More precisely, we prove that every finite solvable cut group of order divisible by 35 has an element of order 35, and that every finite solvable cut group with elements of orders 21 and 35 has an element of order 10. This completes the classification of the Gruenberg–Kegel graphs of finite solvable cut groups.

1. INTRODUCTION

The *Gruenberg–Kegel graph* (or *prime graph*) $\Gamma_{\text{GK}}(G)$ of a finite group G is the graph whose vertices are the primes dividing $|G|$, two distinct primes p, q being adjacent if and only if G has an element of order pq . This graph has been studied intensively since Gruenberg and Kegel showed that the graph of a solvable group is disconnected exactly when the group is Frobenius or 2-Frobenius [13]. A natural problem is to decide which graphs are Gruenberg–Kegel graphs of groups in a given class.

A finite group G is called *cut* if all central units of the integral group ring $\mathbb{Z}G$ are trivial, i.e. of the form $\pm z$ with $z \in Z(G)$. By a classical result (see [12, Corollary 7.1.15]), G is cut if and only if it is *inverse semi-rational*: for every $g \in G$, every generator of $\langle g \rangle$ is conjugate to g or to g^{-1} . Equivalently, every irreducible complex character of G takes values in $\mathbb{Q}$ or in an imaginary quadratic field. Rational groups are cut, and cut groups have received considerable attention in recent years; see [2, 5] and the references there. By a theorem of Bächle [1], the primes dividing the order of a finite solvable cut group lie in $\{2, 3, 5, 7\}$.

The Gruenberg–Kegel graphs of finite solvable cut groups were studied by Bächle, Kiefer, Maheshwary and del Río [2]. They classified those with at most three vertices [2, Theorem A], and proved that a graph with four vertices which is the Gruenberg–Kegel graph of a finite solvable cut group is one of the seven graphs of Figure 1; the graphs (p), (q) and (r) do occur [2, Theorem C]. They asked which of the graphs (s), (t), (u), (v) occur [2, Question E]. Debón, García-Lucas and del Río [5] proved that (s) and (t) do not. (For solvable rational groups the analogous classification was completed in [3, 4].)

In this paper we settle the two remaining cases. Our main results are the following.

Theorem A. *Every finite solvable cut group whose order is divisible by 35 has an element of order 35.*

Theorem B. *Every finite solvable cut group which has elements of order 21 and of order 35 has an element of order 10.*

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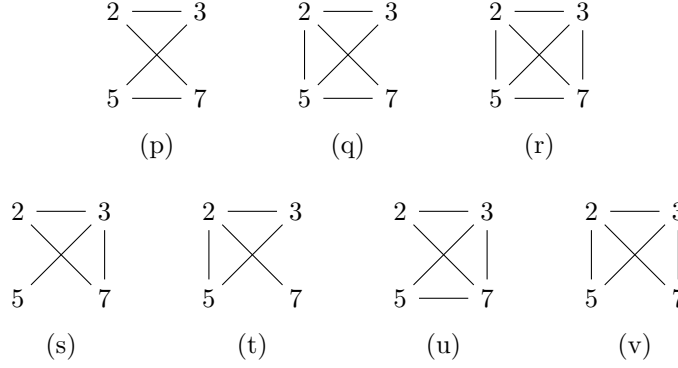


FIGURE 1. The graphs of [2, Theorem C]. The graphs (p), (q), (r) are Gruenberg–Kegel graphs of solvable cut groups; the graphs (s), (t) are not [5]; the graphs (u), (v) are the subject of this paper.

The graphs (s), (t) and (v) are precisely the graphs of Figure 1 without the edge $5 - 7$, and (u) is the only one which contains $3 - 7$ and $5 - 7$ but not $2 - 5$. Hence Theorems A and B give:

Corollary C. *None of the graphs (s), (t), (u), (v) is the Gruenberg–Kegel graph of a finite solvable cut group. Consequently, the Gruenberg–Kegel graph of a finite solvable cut group of order divisible by $2 \cdot 3 \cdot 5 \cdot 7$ is one of the graphs (p), (q), (r) of Figure 1, and together with [2, Theorem A] this completes the classification of the Gruenberg–Kegel graphs of finite solvable cut groups.*

Note that Theorem A gives a new proof of the main result of [5] (that (s) and (t) are not realisable), independent of the main arguments there; in fact no part of this paper relies on the main theorem of [5]. Both hypotheses, solvability and the cut property, are necessary: the solvable groups $(\mathbb{F}_{5^6} \rtimes C_{14}) \times C_3$ and $(\mathbb{F}_{7^4} \rtimes C_{15}) \times C_2$ (with Frobenius actions) have graphs (s) and (t), respectively (compare [8]), and the rational group S_9 has graph (t) (see [5]).

Strategy. Both theorems are proved by considering a counterexample of minimal order. For Theorem A (Section 3) this leads to a group $V \rtimes S$, where V is a faithful irreducible $\mathbb{F}_p S$ -module with $p \in \{5, 7\}$, an element of prime order $r = 12 - p$ of S acts without fixed points on V , every vector of V is an eigenvector with eigenvalue 2 of some element of S , and no element of S acts as the scalar 2. When the Sylow r -subgroup of S is normal, V carries the structure of a vector space over $\mathbb{F}_{p^{r-1}}$; a Galois descent argument together with the theorem of Halasi and Podoski that coprime linear groups have a base of size two [9] produces a vector which is not a 2-eigenvector of any suitable element. The hardest case is $p = 7$ with a non-normal Sylow 5-subgroup $\langle f \rangle$. There we show that $Q_1 = [O_2(S), f]$ acts on each of its homogeneous components through a quotient isomorphic to the extraspecial group 2_-^{1+4} (Lemma 3.8), and combine the Halasi–Podoski theorem with a colouring argument for solvable permutation groups (Lemma 3.9).

For Theorem B (Sections 4 and 5) we first develop some results on solvable cut groups without elements of order 10 acting on modules in characteristic 5 and 7. They show that a faithful irreducible module of this kind produces no elements of order 35 (Theorems 4.3 and 4.7), and hence that a normal Sylow 7-subgroup (resp. 5-subgroup) is centralised by all 5-elements (resp. 7-elements) (Theorem 4.9). A minimal counterexample G then has $F(G) = O_3(G) \times O_7(G)$, and $G/F(G)$ has graph (p); a detailed analysis of the solvable cut groups with graph (p) leads to the final contradiction.

The proofs are free of computer calculations. Computer searches which are consistent with the results, and which guided the work, are described in Appendix A.

AI disclosure and authorship status. I, the person whose name appears between question marks at the page headers of this document, do not claim to be the author of anything, neither the mathematical ideas nor the redaction, in this paper. It was all generated by Claude Opus 5.5 during one single day. I merely provided references on the topic ([2], [3] and [5]) and asked whether the two remainig graphs could be Gruenberg-Kegel graphs of finite solvable cut groups. I have not even read it yet, but diagonally.

2. PRELIMINARIES

2.1. Notation. All groups are finite and all modules are right modules. For a group G , $\pi(G)$ is the set of prime divisors of $|G|$; for a prime p , $O_p(G)$, $O_{p'}(G)$ and $F(G)$ have their usual meaning, and G_p denotes a Sylow p -subgroup. We write $x^g = g^{-1}xg$ and $[x, g] = x^{-1}x^g$. For $g \in G$ we put

$$B_G(g) = N_G(\langle g \rangle) / C_G(g) \leq \text{Aut}\langle g \rangle \cong (\mathbb{Z}/|g|\mathbb{Z})^\times,$$

and we identify $\text{Aut}\langle g \rangle$ with $(\mathbb{Z}/|g|\mathbb{Z})^\times$; when $g = g_1g_2$ with g_1, g_2 commuting elements of coprime orders, we identify $\text{Aut}\langle g \rangle$ with $\text{Aut}\langle g_1 \rangle \times \text{Aut}\langle g_2 \rangle$. An element g is *inverse semi-rational* in G if $B_G(g)\langle -1 \rangle = \text{Aut}\langle g \rangle$, and *rational* if $B_G(g) = \text{Aut}\langle g \rangle$. So G is cut if and only if every element is inverse semi-rational; then $B_G(g)$ has index at most 2 in $\text{Aut}\langle g \rangle$, so it contains all squares of $\text{Aut}\langle g \rangle$, and if the index is 2 then $-1 \notin B_G(g)$. We use these facts constantly. Quotients of cut groups are cut. If V is an $\mathbb{F}_pG$ -module and $\lambda \in \mathbb{F}_p^\times$, we write $V_g(\lambda)$ for the λ -eigenspace of g ; an element “acts as the scalar λ ” if it acts on V as $\lambda \cdot 1_V$. We write E for the extraspecial group 2_-^{1+4} of order 32 and minus type; recall that $\text{Out}(E) \cong O_4^-(2) \cong S_5$.

2.2. Known facts. We collect the facts on cut groups that we use; they are taken from [2] and [5].

Lemma 2.1 ([2, Theorem 2.8, Lemma 2.9]). *If G is a solvable cut group, then $\pi(G) \subseteq \{2, 3, 5, 7\}$; if $5 \in \pi(G)$ then $2 \in \pi(G)$, and if $7 \in \pi(G)$ then $3 \in \pi(G)$.*

Lemma 2.2 ([5, Lemma 3.1]). *Let G be a cut group and $p \neq q$ prime divisors of $|G|$.*

- (1) *If $p \equiv 1 \pmod{4}$, every p -element of G is rational in G . In particular elements of order 5 are rational.*
- (2) *If G_p is abelian, its exponent divides p or 4. In particular a cyclic Sylow p -subgroup, p odd, has order p .*
- (3) *If $G_p \trianglelefteq G$ and G has no element of order pq , then G_q is quaternion of order 8 or cyclic of order dividing q or 4.*
- (4) *If G_2 is cyclic and p is odd, then G has no element of order $4p$.*

Lemma 2.3 ([5, Lemma 3.2(1),(3)]). *Let G be a cut group with $G_2 \cong Q_8$ and let p be an odd prime.*

- (1) *If $a \in G$ has order $4pn$ with $n \geq 1$, then every 2-element of $N_G(\langle a \rangle)$ commutes with the p -part of a .*
- (2) *If V is a normal p -subgroup of G , then $C_{G_2}(V)$ is 1 or G_2 .*

Lemma 2.4. *Let G be a cut group. If G_2 is generalised quaternion, then $G_2 \cong Q_8$.*

Proof. This is the argument of [2, Lemma 3.5]. If $|G_2| = 2^n \geq 16$, let y generate the cyclic subgroup of index 2. Then G_2 normalises $\langle y \rangle$ and induces inversion on it, and $B_G(y)$ is a 2-group whose Sylow 2-subgroup is the image of the Sylow 2-subgroup G_2 of $N_G(\langle y \rangle)$; hence $B_G(y) = \langle -1 \rangle$, while $|\text{Aut}\langle y \rangle| = 2^{n-2} \geq 4$. So y is not inverse semi-rational. $\square$

Lemma 2.5 ([5, Lemma 2.2]). *Let $N \rtimes H$ be a Frobenius group with complement H acting faithfully on a module M over a field whose characteristic does not divide $|NH|$. Then every element of H has a nonzero fixed vector on M .*

We shall also use Lemmas 3.3 and 3.6 of [5], and the following deep result.

Theorem 2.6 (Halasi–Podoski [9]). *Let $H \leq GL(U)$ be a linear group over a finite field of characteristic ℓ with $\ell \nmid |H|$. Then there are $u_1, u_2 \in U$ with $C_H(u_1) \cap C_H(u_2) = 1$.*

2.3. Elementary lemmas.

Lemma 2.7. *Let $G = M \rtimes S$ with $\gcd(|M|, |S|) = 1$, and let $c \in Z(M)$ and $s \in S$ commute. Then $B_G(cs)$ is the image of $N_S(\langle c \rangle) \cap N_S(\langle s \rangle)$ in $\text{Aut}\langle c \rangle \times \text{Aut}\langle s \rangle$. If M is abelian, every element of G is conjugate to such an element cs .*

Proof. Let $x = ms'$ ($m \in M$, $s' \in S$) normalise $\langle cs \rangle$; then x normalises $\langle c \rangle$ and $\langle s \rangle$. If $s \neq 1$, then $s^x = (s[s, m])^{s'} \in S$ forces $[s, m] \in S \cap M = 1$. As $c \in Z(M)$, m centralises cs , and $s' = m^{-1}x$ induces on $\langle cs \rangle$ the same automorphism as x . The last statement is Schur–Zassenhaus. $\square$

Lemma 2.8. *If a cut group has an element of order 105, then it has an element of order 10.*

Proof. Let $|g| = 105$. Then $|B(g)| \geq 24$ inside $\text{Aut}\langle g \rangle \cong C_2 \times C_4 \times C_6$, so the Sylow 2-subgroup of $B(g)$ has order at least 8 in $C_2 \times C_4 \times C_2$. Its projection to $\text{Aut}\langle g_5 \rangle \cong C_4$ therefore has a nontrivial kernel: some 2-element $x \in N(\langle g \rangle) \setminus C(g)$ centralises g_5 , and xg_5 has order divisible by 10. $\square$

In particular, in a cut group without elements of order 10, no nontrivial 3-element centralises two commuting nontrivial elements of orders 5 and 7; we use this repeatedly.

Lemma 2.9. *Let S be a solvable cut group.*

- (a) *If $Z(S)$ contains an element of order 3, then every 3'-element of S is rational; if moreover S_5 is cyclic, then $5 \nmid |S|$.*
- (b) *If $Z(S)$ contains an element of order 4, then every element of odd order of S is rational; if moreover S_7 is cyclic, then $7 \nmid |S|$.*

Proof. (a) If $z \in Z(S)$ has order 3 and s is a 3'-element, then $N(\langle zs \rangle) = N(\langle s \rangle)$ and $C(zs) = C(s)$, so $|B(s)| = |B(zs)| \geq \varphi(3|s|)/2 = \varphi(|s|)$. Suppose $5 \mid |S|$ and S_5 is cyclic, and put $\bar{S} = S/O_{5'}(S)$. Then $O_5(\bar{S}) \neq 1$ is cyclic and $\langle \bar{f} \rangle = \Omega_1(O_5(\bar{S}))$ is normal of order 5. Images of rational elements are rational, so some 2-element $\bar{y}$ (the image of a 2-element of S , hence rational) induces an automorphism α of order 4 of $\langle \bar{f} \rangle$. As $\bar{y}$ is rational, $\bar{y}^{\bar{x}} = \bar{y}^3$ for some $\bar{x}$; since $\text{Aut}\langle \bar{f} \rangle$ is abelian, $\alpha = \alpha^3$, a contradiction. (b) The first statement is proved as in (a), using a central element of order 4. For the second, argue as in (a) with $\bar{S} = S/O_{7'}(S)$, $\langle \bar{a} \rangle = \Omega_1(O_7(\bar{S}))$ of order 7, and a 3-element $\bar{y}$ (rational, being of odd order) inducing an automorphism α of order 3 of $\langle \bar{a} \rangle$: from $\bar{y}^{\bar{x}} = \bar{y}^2$ we get $\alpha = \alpha^2$, a contradiction. $\square$

2.4. Fixed points of elements of prime order. The following observation is standard (compare [10]); we include the short proof in the form we need.

Lemma 2.10. *Let K be a group, $Z \leq Z(K)$ with K/Z abelian, and let b be an automorphism of K centralising Z and acting without fixed points on K/Z . Let $\psi \in \text{Irr}(K)$ be b -invariant and let $\hat{\psi}$ be an extension of ψ to $K \rtimes \langle b \rangle$. Then $|\hat{\psi}(b)| = 1$.*

Proof. As K/Z is abelian and b is fixed-point-free on it, $kZ \mapsto k^{-1}k^{b^{-1}}Z$ is a bijection of K/Z ; hence every element of Kb is K -conjugate to an element of Zb . Since Z is central in $K\langle b \rangle$, $\hat{\psi}(zb) = \mu(z)\hat{\psi}(b)$ for the linear character μ with $\psi_Z = \psi(1)\mu$; so $|\hat{\psi}|$ is constant on Kb . By Gallagher's theorem [11, Corollary 6.17] the characters $\hat{\psi}\lambda$, $\lambda \in \text{Irr}(K\langle b \rangle/K)$, are irreducible and distinct, and their orthogonality relations show that $\sum_{x \in K} |\hat{\psi}(xb^i)|^2$ does not depend on i ; summing over i gives the value $|K|$. Hence $|K| |\hat{\psi}(b)|^2 = |K|$. $\square$

Let K be a q -group and a an automorphism of prime order $r \neq q$ of K . If K is minimal among a -invariant subgroups with $[K, a] \neq 1$, then $K = [K, a]$, $\Phi(K) \leq Z(K)$, a centralises $\Phi(K)$, and a acts irreducibly and without fixed points on $K/\Phi(K)$ [7, 5.3.7]; in particular $|K/\Phi(K)| = q^{\text{ord}_r(q)}$.

Lemma 2.11. *Let S act faithfully on a vector space V over a field of characteristic ℓ , let $a \in S$ have prime order r and act without fixed points on V , and let $K \leq S$ be a q -subgroup, $q \notin \{\ell, r\}$, which is minimal among a -invariant subgroups with $[K, a] \neq 1$. Then $\text{ord}_r(q) = 2n$ is even and $q^n \leq r - 1$. Consequently, if $(r, q) \in \{(7, 2), (7, 3), (5, 3), (3, 5)\}$, then a centralises every a -invariant q -subgroup of S .*

Proof. We may assume the field algebraically closed; then $\ell \neq r$, since a is fixed-point-free. Write $V = \bigoplus_{\mu} V_{\mu}$ according to the linear characters μ of $\Phi(K)$; each V_{μ} is $K\langle a \rangle$ -invariant. For each μ , $\beta_{\mu}(x\Phi(K), y\Phi(K)) = \mu([x, y])$ is an a -invariant alternating form on $K/\Phi(K)$ (commutators lie in $\Phi(K) \leq Z(K)$ and have order q), so by irreducibility its radical is 0 or $K/\Phi(K)$. If it is $K/\Phi(K)$, the image $\tilde{K}$ of K in $GL(V_{\mu})$ is abelian; if $[\tilde{K}, a] \neq 1$, then $\Omega_1([\tilde{K}, a]) \rtimes \langle a \rangle$ is a Frobenius group acting faithfully on V_{μ} , and a has a fixed point by Lemma 2.5; so a centralises $\tilde{K}$, and as $K = [K, a]$, K acts trivially on V_{μ} . If β_{μ} is nondegenerate, $|K/\Phi(K)| = q^{2n}$ and there is a unique $\chi \in \text{Irr}(K \mid \mu)$, of degree q^n ; it is a -invariant and extends to $K\langle a \rangle$, and every $K\langle a \rangle$ -composition factor of V_{μ} affords an extension $\hat{\chi}$ (a Brauer character, which is an ordinary character as $\ell \nmid |K\langle a \rangle|$). By Lemma 2.10 (applied to a^k , $r \nmid k$), the multiplicity of the eigenvalue 1 of a on it is $\frac{1}{r} \sum_{k=0}^{r-1} \hat{\chi}(a^k) \geq (q^n - (r - 1))/r$. As $K \neq 1$ acts faithfully, some V_{μ} is of the second kind, and $q^n \leq r - 1$. For the listed pairs, $q^{\text{ord}_r(q)/2} > r - 1$ or $\text{ord}_r(q)$ is odd: $\text{ord}_7(2) = 3$, $3^{6/2} > 6$, $3^{4/2} > 4$, $5^{2/2} > 2$. If a acted nontrivially on an a -invariant q -subgroup P , a subgroup $K \leq P$ minimal among the a -invariant subgroups with $[K, a] \neq 1$ would contradict the first part. $\square$

3. PROOF OF THEOREM A

3.1. The minimal configuration.

Proposition 3.1. *Let G be a solvable cut group of minimal order such that 35 divides $|G|$ and G has no element of order 35. Then $G = V \rtimes S$, where for some $p \in \{5, 7\}$ and $r = 12 - p$:*

- (i) $V = F(G)$ is an elementary abelian Sylow p -subgroup of G and a faithful irreducible $\mathbb{F}_p S$ -module;
- (ii) S is a solvable cut $\{2, 3, r\}$ -group, and its Sylow r -subgroup has order r and acts without fixed points on V ;
- (iii) for every $v \in V$ there is $s \in S$ with $vs = 2v$, and s can be taken to be a 2-element if $p = 5$ and a 3-element if $p = 7$;
- (iv) no element of S acts on V as the scalar 2.

Proof. Let N be a minimal normal subgroup of G . If 35 divided $|G/N|$, then by minimality G/N would have an element of order 35, and so would G . Hence N is a Sylow 5- or Sylow 7-subgroup of G . A minimal normal Sylow 5-subgroup and a minimal normal Sylow 7-subgroup would commute, so there is $p \in \{5, 7\}$ such that every minimal normal subgroup is a Sylow p -subgroup; thus there is exactly one, V , and $F(G) = V$. By Schur–Zassenhaus $G = V \rtimes S$, and $C_S(V) = 1$ as $C_G(F(G)) \leq F(G)$. The r -elements of S have no fixed points on V , so a Sylow r -subgroup of S is a Frobenius complement of odd order, hence cyclic, hence of order r by Lemma 2.2(2); and $\pi(S) \subseteq \{2, 3, r\}$ by Lemma 2.1. For (iii): if $p = 5$, v is rational by Lemma 2.2(1); if $p = 7$, $B_G(v)$ has order at least 3, hence contains $\{1, 2, 4\}$. By Lemma 2.7, $B_G(v)$ is induced by the stabiliser of $\mathbb{F}_p v$ in S , and a suitable power of an element inducing $v \mapsto 2v$ is a 2-element (resp. 3-element) still inducing $v \mapsto 2v$. For (iv): an element acting as the scalar 2 is central in S (as V is faithful)

and has order 4 if $p = 5$ and order 3 if $p = 7$. This contradicts Lemma 2.9(b) if $p = 5$ (as S_7 has order 7) and Lemma 2.9(a) if $p = 7$ (as S_5 has order 5). $\square$

From now on in this section $G = V \rtimes S$ is as in Proposition 3.1. We first record a consequence of Lemma 2.11.

Lemma 3.2. *If $p = 5$, the Sylow 7-subgroup $\langle a \rangle$ of S is normal in S .*

Proof. Let $D = O_{7'}(S)$. As $O_{7'}(S/D) = 1$, $F(S/D)$ is a 7-group, so $M = \langle a \rangle D \trianglelefteq S$. If $[D, a] = 1$, then $\langle a \rangle$ is characteristic in $M = \langle a \rangle \times D$, hence normal in S . Otherwise a acts nontrivially on an a -invariant Sylow q -subgroup of D for some $q \in \{2, 3\}$, contradicting Lemma 2.11 with $r = 7$. $\square$

3.2. Galois descent. Call a group H *half-rational* if $|B_H(c)| \geq \frac{1}{2}|\text{Aut}\langle c \rangle|$ for every $c \in H$.

Lemma 3.3. *Let H be a half-rational group, ℓ a prime not dividing $|H|$, E a finite field of characteristic ℓ containing $\mathbb{F}_{\ell^2}$, and V an EH -module. Then $V \cong V_1 \otimes_{\mathbb{F}_{\ell^2}} E$ for some $\mathbb{F}_{\ell^2}H$ -module V_1 , with H acting on the first factor.*

Proof. Let $q = \ell^2$ and let X be an absolutely irreducible representation of H occurring in $V \otimes_E \bar{E}$. For $c \in H$, $B_H(c)$ has index at most 2 in $(\mathbb{Z}/|c|\mathbb{Z})^\times$, so it contains the square q ; thus c^q is conjugate to c , and $\text{tr } X(c)^q = \text{tr } X(c^q) = \text{tr } X(c)$. So the traces of X lie in $\mathbb{F}_q$, and since the Schur indices over finite fields are trivial, X is realisable over $\mathbb{F}_q$, say by X_0 . Then $X_0 \otimes_{\mathbb{F}_q} E$ is absolutely irreducible, and every irreducible EH -module is of this form. As V is semisimple, the lemma follows. $\square$

Lemma 3.4. *Let $r \in \{5, 7\}$ and suppose the Sylow r -subgroup $\langle a \rangle$ of S is normal. Let $C = C_S(a) = \langle a \rangle \times C'$ with $C' = O_{r'}(C)$, and $E = \mathbb{F}_p[a] \subseteq \text{End}(V)$. Then $E \cong \mathbb{F}_{p^{r-1}}$, V is an E -vector space on which C' acts E -linearly, C' is half-rational, and $V \cong V_1 \otimes_{\mathbb{F}_{p^2}} E$ as C' -modules.*

Proof. As $\text{ord}_7(5) = 6$ and $\text{ord}_5(7) = 4$, the r -th cyclotomic polynomial is irreducible over $\mathbb{F}_p$ and, a being fixed-point-free, $E \cong \mathbb{F}_{p^{r-1}}$. For $c \in C'$ we have $N_S(\langle ca \rangle) = N_S(\langle c \rangle) \cap N_S(\langle a \rangle)$, and the elements of $B_S(ca)$ acting trivially on $\langle a \rangle$ form a subgroup of $B_C(c) = B_{C'}(c)$. Since S is cut, $|B_S(ca)| \geq \frac{1}{2}|\text{Aut}\langle c \rangle|(r-1)$, while the image of $B_S(ca)$ in $\text{Aut}\langle a \rangle$ has order at most $r-1$; hence $|B_{C'}(c)| \geq \frac{1}{2}|\text{Aut}\langle c \rangle|$. Now apply Lemma 3.3. $\square$

3.3. The case $p = 7$ with a normal Sylow 5-subgroup.

Proposition 3.5. *If $p = 7$, the Sylow 5-subgroup of S is not normal in S .*

Proof. Suppose $F = \langle f \rangle \trianglelefteq S$ and use Lemma 3.4 with $a = f$: $E = \mathbb{F}_{7^4}$ and $V \cong V_1 \otimes_{\mathbb{F}_{49}} E$ as C' -modules. As S/C embeds in $\text{Aut } F \cong C_4$, all 3-elements of S lie in C' . Let $H \leq GL(V_1)$ be the group generated by the image of C' and the scalar 4; it is a $7'$ -group. Take $v_1, v_2 \in V_1$ as in Theorem 2.6 and $e \in E \setminus \mathbb{F}_{49}$, and put $w = v_1 \otimes 1 + v_2 \otimes e$. By Proposition 3.1(iii) there is a 3-element $t \in C'$ with $wt = 2w$. Writing $t = t_1 \otimes 1$, and comparing components in $V_1 \otimes 1 \oplus V_1 \otimes e$, we get $v_i t_1 = 2v_i$ ($i = 1, 2$). So $4t_1 \in C_H(v_1) \cap C_H(v_2) = 1$, and t acts on V as the scalar 2, contrary to Proposition 3.1(iv). $\square$

3.4. The case $p = 5$.

Proposition 3.6. *In Proposition 3.1, $p \neq 5$.*

Proof. Let $p = 5$. By Lemma 3.2, $\langle a \rangle \trianglelefteq S$, and we use Lemma 3.4: $E = \mathbb{F}_{5^6}$, $V \cong V_1 \otimes_{\mathbb{F}_{25}} E$ as C' -modules, $m := \dim_E V = \dim_{\mathbb{F}_{25}} V_1$. Let $\zeta \in E$ be the element by which a acts, $E_0 = \mathbb{F}_{125} \subseteq E$, and σ the automorphism of order 2 of E/E_0 , so $\zeta^\sigma = \zeta^{125} = \zeta^{-1}$. As a is inverse semi-rational, $S/C \leq \text{Aut}\langle a \rangle \cong C_6$ contains the subgroup of order 3. Put $H^* = \langle C', 3 \rangle$ (with 3 the scalar),

which acts E -linearly on V , and let $H \leq GL(V_1)$ be its image. Fix $e \in E \setminus \mathbb{F}_{25}$, so that $1, e, e^2$ is an $\mathbb{F}_{25}$ -basis of E .

Step 1: a line with trivial stabilisers. If $m = 1$, H acts on V_1 by scalars; let $w = v_1 \otimes 1$ with $0 \neq v_1 \in V_1$. If $m \geq 2$, Theorem 2.6 gives $v_1, v_2 \in V_1$ with $C_H(v_1) \cap C_H(v_2) = 1$, which we may take linearly independent (if, say, $v_2 \in \mathbb{F}_{25}v_1$, then $C_H(v_1) = 1$ and we may replace v_2 by any vector outside $\mathbb{F}_{25}v_1$; similarly with the roles exchanged); let $w = v_1 \otimes 1 + v_2 \otimes e$. Put $L = Ew$. If $h \in H^*$ fixes a nonzero vector of L , it fixes w (it is E -linear), hence v_1 and v_2 , so $h = 1$.

Step 2: elements covering L . Let $0 \neq x \in L$, and let $y \in S$ be a 2-element with $xy = 2x$ (Proposition 3.1(iii)). If $y \in C$, then $y \in C'$ and $3y \in H^*$ fixes x , so y acts as the scalar 2, contrary to (iv). Hence $S/C \cong C_6$, y maps to its involution, $a^y = a^{-1}$, and y is σ -semilinear on V . Moreover $y^2 \in C'$ and $xy^2 = 4x$, so $-y^2 \in H^*$ fixes x , and $y^2 = -1$. Thus $\varphi = 3y$ is a σ -semilinear involution fixing x , and $(\lambda x)\varphi = \lambda^\sigma x$ for $\lambda \in E$; in particular φ stabilises L .

Step 3: the stabiliser of L . Let $\mathcal{J}$ be the set of all $\varphi = 3y$, $y \in S$, which are involutions with a nonzero fixed vector $x \in L$. For such φ , $\varphi^2 = 1$ gives $y^2 = -1$, so y is a 2-element with $xy = 2x$, and by Step 2 φ is σ -semilinear; moreover, by Step 2 every nonzero vector of L is fixed by some $\varphi \in \mathcal{J}$. Let St be the stabiliser of L in $\langle a \rangle \times H^*$. Each $g \in \text{St}$ acts on the 1-dimensional E -space L as a scalar $\hat{g} \in E^\times$, and $g \mapsto \hat{g}$ is an injective homomorphism: if $g = a^i h$ ($h \in H^*$) acts trivially on L , then a power a^{ik} of g with $7 \nmid k$ acts trivially on L , so $7 \mid i$, and $h = 1$ by Step 1. Fix $\varphi_1 \in \mathcal{J}$ and $0 \neq x_1 \in L$ fixed by φ_1 . For $\varphi \in \mathcal{J}$, $g = \varphi_1 \varphi$ is E -linear and lies in St , and $(\lambda x_1)\varphi = (\lambda x_1)\varphi_1 g = \lambda^\sigma \hat{g} x_1$. Hence the fixed vectors of φ in L are the λx_1 with $\lambda^{124} = \hat{g}^{-1}$. As $\lambda \mapsto \lambda^{124}$ maps $E^\times$ onto the group of 126-th roots of unity with kernel $E_0^\times$, covering $L \setminus \{0\}$ requires that the $\hat{g}$ fill all 126-th roots of unity. So St contains an element g with $\hat{g}$ of order 126; the 7'-part of g lies in H^* , and its 3-part s lies in C' (as H^*/C' is a 2-group), stabilises L and acts on it as a scalar μ of order 9.

Step 4: contradiction. If $m = 1$, s acts on V_1 as an element of $\mathbb{F}_{25}^\times$, so $\mu \in \mathbb{F}_{25}^\times$. If $m \geq 2$, write $s = s_1 \otimes 1$, $\mu = \mu_0 + \mu_1 e + \mu_2 e^2$ and $e^3 = c_0 + c_1 e + c_2 e^2$ with $\mu_i, c_i \in \mathbb{F}_{25}$. Comparing the components in $V_1 \otimes e^2$ of $ws = v_1 s_1 \otimes 1 + v_2 s_1 \otimes e$ and of $\mu w = v_1 \otimes \mu + v_2 \otimes \mu e$ gives $0 = \mu_2 v_1 + (\mu_1 + \mu_2 c_2) v_2$, so $\mu_1 = \mu_2 = 0$ and again $\mu \in \mathbb{F}_{25}^\times$. But $9 \nmid 24$. $\square$

3.5. The case $p = 7$ with a non-normal Sylow 5-subgroup. In this subsection $p = 7$, $F = \langle f \rangle$ is a Sylow 5-subgroup of S , which by Proposition 3.5 is not normal, $Q = O_2(S)$ and $Q_1 = [Q, f]$.

Lemma 3.7. (a) $QF \trianglelefteq S$, so $S = Q N_S(F)$;

(b) $Q_1 \trianglelefteq S$ and $Q_1 \neq 1$;

(c) every 3-element of S is of the form t^x with $t \in C_S(f)$ a 3-element and $x \in Q_1$.

Proof. (a) Let $D = O_{5'}(S)$, a $\{2, 3\}$ -group. As $O_{5'}(S/D) = 1$, $F(S/D) = FD/D$, so $FD \trianglelefteq S$. By Lemma 2.11 (with $r = 5$, $q = 3$), f centralises every f -invariant 3-subgroup of S . We have $Q = O_2(D)$. Put $\bar{D} = D/Q$; then $O_2(\bar{D}) = 1$, so $C_{\bar{D}}(O_3(\bar{D})) \leq O_3(\bar{D})$. As $O_3(\bar{D})$ is the image of an f -invariant Sylow 3-subgroup of $O_{2,3}(D)$, f centralises $O_3(\bar{D})$. In $\bar{D}\langle f \rangle$ we have $[O_3(\bar{D}), f, \bar{D}] = 1 = [\bar{D}, O_3(\bar{D}), f]$, so by the three subgroups lemma $[\bar{D}, f]$ centralises $O_3(\bar{D})$, i.e. $[\bar{D}, f] \leq O_3(\bar{D})$; then $[\bar{D}, f] = [\bar{D}, f, f] = 1$. Hence $FD/Q = FQ/Q \times D/Q$, its normal Sylow 5-subgroup FQ/Q is characteristic, $FQ \trianglelefteq S$, and the Frattini argument gives $S = Q N_S(F)$.

(b) Q_1 is normalised by Q , F and $N_S(F)$, hence by S . If $Q_1 = 1$, then $[D, f] = [D, f, f] \leq [Q, f] = 1$ (as $[D, f] \leq Q$ by the proof of (a)), so F is characteristic in $FD = F \times D \trianglelefteq S$, and $F \trianglelefteq S$, a contradiction.

(c) By (a), a Sylow 3-subgroup P of $N_S(F)$ is a Sylow 3-subgroup of S ; it centralises F , as $\text{Aut } F$ is a 2-group. A 3-element of S lies in P^{ny} for some $n \in N_S(F)$ and $y \in Q$, and $P^n \leq C_S(F^n) = C_S(f)$. As $Q = C_Q(f)Q_1$ (coprime action), $y = cx$ with $c \in C_Q(f)$, $x \in Q_1$, and $P^{nc} \leq C_S(f)$. $\square$

Lemma 3.8. *Let X be a 2-group, f an automorphism of X with $f^5 = 1$, and Δ a finite-dimensional module for $X \rtimes \langle f \rangle$ over an algebraically closed field k of characteristic 7 on which f has no eigenvalue 1. Then every nontrivial irreducible constituent φ of $\Delta|_{[X,f]}$ satisfies $[X, f]/\ker \varphi \cong E$.*

Proof. Induction on $|X|$. If f acts trivially on X there is nothing to prove; let f have order 5. We may assume Δ irreducible. If a constituent θ of $\Delta|_X$ is f -invariant, it extends to $X\langle f \rangle$, and by Gallagher's theorem $\Delta|_X$ is irreducible. Otherwise Δ is induced from X , and f permutes five summands cyclically and has the eigenvalue 1. So $\Delta|_X$ is irreducible with f -invariant character θ . Replacing X by $X/\ker \theta$ (which is harmless, as $[X, f]$ maps onto $[X/\ker \theta, f]$ with kernel contained in the kernel of every constituent of $\theta|_{[X,f]}$), we may assume that θ is faithful. Then $Z = Z(X)$ is cyclic and centralised by f . If X is abelian, then $X = Z$ and $[X, f] = 1$. Otherwise let $Z < A \leq X$ with A/Z a minimal normal subgroup of $X\langle f \rangle/Z$. As X acts trivially on this irreducible module in characteristic 2, $[A, X] \leq Z$, and A/Z is an irreducible $\mathbb{F}_2\langle f \rangle$ -module: either $|A/Z| = 2$, or $|A/Z| = 16$ and f is fixed-point-free on A/Z .

Case $|A/Z| = 2$. Then A is abelian, and θ_A is not homogeneous (otherwise A acts by scalars and $A \leq Z$, as θ is faithful). So θ_A has two constituents, both f -invariant, with common stabiliser I of index 2 in X ; I is f -invariant. The corresponding homogeneous component Δ_0 of $\Delta|_A$ is an f -invariant subspace, hence an $I\langle f \rangle$ -module without the eigenvalue 1 for f , and $\Delta|_I = \Delta_0 \oplus \Delta_0 u$ for $u \in X \setminus I$. As X/I is an f -trivial quotient, $[X, f] = [I, f]$, which is normal in $X\langle f \rangle$. By induction the nontrivial constituents of $\Delta_0|_{[I,f]}$ have the required kernels, and those of $\Delta_0 u|_{[I,f]}$ are their u -conjugates.

Case $|A/Z| = 16$. For $a, b \in A$, $[a, b] \in Z$ and $[a, b]^2 = [a^2, b] = 1$, so $(aZ, bZ) \mapsto [a, b]$ is an f -invariant alternating form on A/Z with values in $\Omega_1(Z) \cong C_2$, whose radical is 0 or A/Z . Suppose it is A/Z , so A is abelian, and let λ be a constituent of θ_A . Then $\lambda^x = \lambda \nu_x$ with $\nu_x(a) = \lambda([a, x])$, and $x \mapsto \nu_x$ is a homomorphism $X \rightarrow \text{Irr}(A/Z)$; so the set Λ of constituents of θ_A is a coset of a subgroup Ψ of $\text{Irr}(A/Z)$, namely $\Psi = \{\nu : \Lambda\nu = \Lambda\}$, which is f -invariant. As $\text{Irr}(A/Z)$ is an irreducible f -module, either $\Psi = 1$, in which case θ_A is homogeneous and $A \leq Z$, impossible; or $|\Lambda| = 16$, in which case f acts on Λ as an affine map with fixed-point-free linear part, with one fixed point and three orbits of length 5. For λ_1 in such an orbit and $0 \neq v$ in the λ_1 -component of Δ , $v + vf + \dots + vf^4 \neq 0$ is fixed by f , a contradiction. Hence the form is nondegenerate.

Put $E_A = [A, f]$. Since $A/Z = [A/Z, f]$, $A = E_A Z$. As $E_A = [E_A, f]$, f is fixed-point-free on E_A/E'_A , so $E_A \cap Z \leq E'_A \leq \Omega_1(Z)$; and E_A is nonabelian, since A is. So $E_A \cap Z = E'_A$ has order 2, $|E_A| = 32$, $E_A/E'_A \cong A/Z$ carries a nondegenerate commutator form, and E_A is extraspecial; it is of minus type because it admits an automorphism of order 5 ($5 \nmid |O_4^+(2)|$). Thus $E_A \cong E$.

Each $x \in X$ acts on A as $a \mapsto a\phi_x(a)$ with $\phi_x \in \text{Hom}(A/Z, \Omega_1(Z))$, and by nondegeneracy $\phi_x = [\cdot, e]$ for some $e \in E_A$. Hence $X = E_A D$ with $D = C_X(A)$, an f -invariant subgroup with $[E_A, D] = 1$ and $E_A \cap D = Z(E_A)$. Thus $X = E_A \circ D$, $\theta = \psi \cdot \delta$ with ψ the faithful irreducible character of E_A (of degree 4) and $\delta \in \text{Irr}(D)$ faithful and f -invariant, and $\Delta = M \otimes M'$ accordingly. As f normalises E_A and D and fixes ψ and δ , it acts as $a \otimes b$ with $a^{-1}\rho_\psi(e)a = \rho_\psi(e^f)$ and $b^{-1}\rho_\delta(d)b = \rho_\delta(d^f)$, and after rescaling $a^5 = 1 = b^5$. Then $ef^j \mapsto \rho_\psi(e)a^j$ is a representation of $E_A \rtimes \langle f \rangle$ extending ψ , whose (Brauer) character $\hat{\psi}$ satisfies $|\hat{\psi}(f^j)| = 1$ for $5 \nmid j$ by Lemma 2.10. If m_i is the multiplicity of the eigenvalue ζ^i of a (ζ a primitive fifth root of unity), then $5 \sum_i m_i^2 = \sum_{j=0}^4 |\hat{\psi}(f^j)|^2 = 16 + 4$, so a has four distinct eigenvalues ζ^i , $i \neq i_0$. As $a \otimes b$ has no eigenvalue 1, every eigenvalue of b is ζ^{-i_0} , so b is a scalar and f centralises $\rho_\delta(D)$; as δ is faithful, $[D, f] = 1$. Therefore $[X, f] = E_A$, and $\theta|_{E_A} = \delta(1)\psi$ with ψ faithful. $\square$

Lemma 3.9. *Let P be a solvable $\{2, 3\}$ -group acting faithfully on a finite set Ω , and let $c \geq 6$. Then there is a map $\chi : \Omega \rightarrow \{1, \dots, c\}$ such that no element of order 3 of P preserves χ (i.e. satisfies $\chi(\omega g) = \chi(\omega)$ for all $\omega \in \Omega$).*

Proof. Call such a map *good*. If P has orbits $\Omega_1, \dots, \Omega_s$, a map whose restriction to each Ω_i is good for the group induced by P on Ω_i is good. So let P be transitive. We show by induction on $n = |\Omega|$ that, for every $c \geq 6$, the good maps form at least c orbits under P ; for $n = 1$ this is clear.

P primitive. Then $P = V \rtimes H$ is affine, with V regular of order $n = p^d$, $p \in \{2, 3\}$, and $H \leq GL(d, p)$ an irreducible $\{2, 3\}$ -group. A nonidentity element fixes the points of a proper affine subspace, hence at most n/p points; so an element of order 3 has at most $\lfloor en \rfloor$ cycles, where $e = (1 + 2/p)/3$, and preserves at most $c^{\lfloor en \rfloor}$ maps. If $n = 2$, $P = C_2$ and all c^2 maps are good, forming $c(c+1)/2$ orbits. If $n = 3$, an element of order 3 preserves only the constant maps: there are at least $(c^3 - c)/6 \geq c$ orbits of good maps. If $n = 4$, $P \in \{A_4, S_4\}$ has eight elements of order 3, each with two cycles: at least $(c^4 - 8c^2)/24 \geq c$ orbits. The case $n = 8$ does not occur, since 7 divides the order of every irreducible subgroup of $GL(3, 2)$. In all other cases $2|P| \leq c^{n - \lfloor en \rfloor}$: for $n = 9, 16, 27, 32$ use $|H| \leq |GL(d, p)|_{\{2, 3\}} = 48, 576, 864, 9216$ respectively ($6^4 \geq 2 \cdot 9 \cdot 48$, $6^6 \geq 2 \cdot 16 \cdot 576$, $6^{12} \geq 2 \cdot 27 \cdot 864$, $6^{11} \geq 2 \cdot 32 \cdot 9216$), and for $n = 2^d$ with $d \geq 6$, or $n = 3^d$ with $d \geq 4$, use $|P| < p^{d^2+d}$ and $(1 - e)n \log_2 6 \geq (d^2 + d + 1) \log_2 p$. Then at least $c^n - |P|c^{\lfloor en \rfloor} \geq c^n/2$ maps are good, and they form at least $c^n/(2|P|) \geq c^{\lfloor en \rfloor} \geq c$ orbits.

P imprimitive. Choose a system of blocks $B_1, \dots, B_m$, $1 < m < n$, on which P acts primitively, with image $\bar{P}$, and let P_1 be the (transitive solvable $\{2, 3\}$ -) group induced on B_1 by its stabiliser. Fix $g_i \in P$ with $B_1 g_i = B_i$. The *type* of a map $\psi : B_i \rightarrow \{1, \dots, c\}$ is the P_1 -orbit of $\omega \mapsto \psi(\omega g_i)$, $\omega \in B_i$; it does not depend on the choice of g_i , and if $g \in P$ preserves a map χ on Ω and $B_i g = B_j$, then $\chi|_{B_i}$ and $\chi|_{B_j}$ have the same type. By induction there are $N \geq c$ types of good maps on B_1 ; choose representatives $\psi_1, \dots, \psi_N$. For a good map $\sigma : \{1, \dots, m\} \rightarrow \{1, \dots, N\}$ for $\bar{P}$ (the primitive case gives at least N orbits of them), let χ_σ be the map given on B_i by $\omega g_i \mapsto \psi_{\sigma(i)}(\omega)$. If $g \in P$ has order 3 and preserves χ_σ , its image in $\bar{P}$ preserves σ , hence is trivial; so g stabilises each B_i and induces there, after transport by g_i , an element of P_1 of order dividing 3 which preserves the good map $\psi_{\sigma(i)}$, hence acts trivially. So $g = 1$, χ_σ is good, and maps $\chi_\sigma, \chi_{\sigma'}$ with σ, σ' in different $\bar{P}$ -orbits lie in different P -orbits. $\square$

Proposition 3.10. *In Proposition 3.1, $p \neq 7$.*

Proof. Let $p = 7$. By Proposition 3.5, $F = \langle f \rangle$ is not normal in S , and we use the notation of Lemma 3.7. Let $k = \mathbb{F}_7$.

Step 1: the homogeneous components of $V|_{Q_1}$. By Lemma 3.8 (with $X = Q$ and $\Delta = V \otimes k$), every irreducible constituent of $(V \otimes k)|_{Q_1}$ is trivial or has kernel with quotient isomorphic to E . The fixed space V^{Q_1} is S -invariant and proper, hence 0. So every constituent θ is the faithful character of degree 4 of $Q_1/\ker \theta \cong E$; its values $0, \pm 4$ are rational, so θ is afforded by an absolutely irreducible $\mathbb{F}_7 Q_1$ -module M_θ , with representation ρ_θ . Let Θ be the set of these characters and V_θ the M_θ -homogeneous component of $V|_{Q_1}$. Then $V = \bigoplus_{\theta \in \Theta} V_\theta$, S permutes the V_θ transitively, and $V_\theta = M_\theta \otimes U_\theta$ with Q_1 acting on the first factor; $\kappa = \dim U_\theta$ does not depend on θ . Each θ is f -invariant: otherwise some irreducible $k[Q_1 F]$ -submodule of $V \otimes k$ is induced from Q_1 , and f has the eigenvalue 1 on it. So F , and hence $\langle f^S \rangle$, acts trivially on Θ , and the group $\hat{S}$ induced by S on Θ is a solvable $\{2, 3\}$ -group. Let N_1 be the kernel of this action.

Step 2: tensor form. If $s \in S$ maps V_θ onto $V_{\theta'}$, then s is a Q_1 -isomorphism from V_θ to the s -twist of $V_{\theta'}$, so $s|_{V_\theta} = a_s \otimes b_s$ with $a_s : M_\theta \rightarrow M_{\theta'}$, $b_s : U_\theta \rightarrow U_{\theta'}$ linear and unique up to reciprocal scalars; since $xs = sx^s$ for $x \in Q_1$, we get $\rho_\theta(x)a_s = a_s \rho_{\theta'}(x^s)$. Let $s \in S$ be a 3-element fixing θ . The image of the stabiliser of θ in $\text{Out}(Q_1/\ker \theta) \cong S_5$ is solvable and contains the image of

f , which has order 5 (as $Q_1/\ker \theta = [Q_1/\ker \theta, f]$, f induces a non-inner automorphism); the solvable subgroups of S_5 of order divisible by 5 lie in a Frobenius group of order 20. So s induces an inner automorphism of $Q_1/\ker \theta$, of 3-power order, hence the trivial automorphism. Therefore a_s commutes with $\rho_\theta(Q_1)$, a_s is a scalar, and s acts on V_θ as $1 \otimes h$. Let C_0 be the subgroup of S of all elements acting on every V_θ as $1 \otimes h_\theta(s)$, and $U = \bigoplus_\theta U_\theta$. Then $h : C_0 \rightarrow GL(U)$ is an injective homomorphism, and every 3-element of N_1 lies in C_0 .

Step 3: coordinates. Let M be the faithful irreducible $\mathbb{F}_7\mathbf{E}$ -module of dimension 4, with $\mathbf{E} \leq GL(M)$, and $N = N_{GL(M)}(\mathbf{E})$. Every automorphism of $\mathbf{E}$ is induced by N , and $N/\mathbb{F}_7^\times \mathbf{E} \cong \text{Out}(\mathbf{E}) \cong S_5$. Fix $f_0 \in N$ of order 5. For each θ let a_θ be the M -part of f on V_θ , normalised so that $a_\theta^5 = 1$ (possible as a_θ^5 commutes with $\rho_\theta(Q_1)$, so is a scalar, and fifth roots are unique in $\mathbb{F}_7^\times$). The elements of order 5 of $\text{Aut}(\mathbf{E}) \cong 2^4.S_5$ are all conjugate (their images are 5-cycles, and two elements of order 5 in the same coset of 2^4 in $2^4 \rtimes C_5$ are conjugate by Sylow's theorem). Hence there are an isomorphism $\alpha : Q_1/\ker \theta \rightarrow \mathbf{E}$ carrying the automorphism induced by f to conjugation by f_0 , and an isomorphism $\iota_\theta : M_\theta \rightarrow M$ with $\iota_\theta^{-1} \rho_\theta(x) \iota_\theta = \alpha(x \ker \theta)$ for $x \in Q_1$. Then $\iota_\theta^{-1} a_\theta \iota_\theta$ and f_0 induce the same automorphism of $\mathbf{E}$, so they differ by a scalar, and both have order 5; so $\iota_\theta^{-1} a_\theta \iota_\theta = f_0$. Put

$$\mu = C_N(f_0) \quad \text{and} \quad Y = \mathbb{F}_7^\times \mathbf{E} \langle f_0 \rangle, \quad |Y| = 480.$$

Then $\mu = \mathbb{F}_7^\times \times \langle f_0 \rangle \leq Y$: if $n \in N$ commutes with f_0 , its image in S_5 centralises a 5-cycle, so $n = cf_0^j$ with $c = \lambda e \in \mathbb{F}_7^\times \mathbf{E}$ commuting with f_0 ; as f_0 is fixed-point-free on $\mathbf{E}/Z(\mathbf{E})$, $e \in Z(\mathbf{E}) = \{\pm 1\}$.

Claim A: if $t \in C_S(f)$ maps V_θ onto $V_{\theta'}$, then $\hat{a}_t := \iota_{\theta'}^{-1} a_t \iota_{\theta'} \in \mu$. Indeed, $ft = tf$ gives $a_\theta a_t = \lambda a_t a_{\theta'}$ for some $\lambda \in \mathbb{F}_7^\times$, so $\hat{a}_t^{-1} f_0 \hat{a}_t = \lambda f_0$, and $\lambda = 1$ by comparing orders. By Step 2, $\hat{a}_t^{-1} (\iota_\theta^{-1} \rho_\theta(x) \iota_\theta) \hat{a}_t = \iota_{\theta'}^{-1} \rho_{\theta'}(x^t) \iota_{\theta'} \in \mathbf{E}$ for $x \in Q_1$, so $\hat{a}_t \in N$, and $\hat{a}_t \in C_N(f_0) = \mu$.

Claim B: for $m \in \{1, 2\}$, Y has at least 6 orbits on the set $\mathcal{G}_m$ of m -dimensional subspaces of M . The group Y acts on $\mathcal{G}_m$ through $Y/\mathbb{F}_7^\times$, of order 80. Now $|\mathcal{G}_2| = 2850 > 35 \cdot 80$, and $|\mathcal{G}_1| = 400 = 5 \cdot 80$, while a non-central involution of $\mathbf{E}$ (with eigenvalues 1 and -1 , each of multiplicity 2) fixes 16 points of $\mathcal{G}_1$; so not all orbits on $\mathcal{G}_1$ are regular and there are at least 6.

Step 4: the vector. Let $H \leq GL(U)$ be generated by $h(C_0)$ and the scalar 4, a $7'$ -group, and choose $u_1, u_2 \in U$ with $C_H(u_1) \cap C_H(u_2) = 1$ (Theorem 2.6). Let $m = \min(2, \kappa)$. For each θ choose linearly independent $y_1^\theta, \dots, y_m^\theta \in U_\theta$ whose span contains the U_θ -components of u_1 and u_2 . By Lemma 3.9, applied to $\hat{S}$ acting on Θ with the Y -orbits on $\mathcal{G}_m$ as colours (Claim B), there is a map χ from Θ to the set of Y -orbits on $\mathcal{G}_m$ which is preserved by no element of order 3 of $\hat{S}$. For each θ choose $P_\theta \in \mathcal{G}_m(M_\theta)$ with $P_\theta \iota_\theta \in \chi(\theta)$, with basis $e_1^\theta, \dots, e_m^\theta$, and put

$$w = \sum_{\theta \in \Theta} w_\theta, \quad w_\theta = \sum_{j=1}^m e_j^\theta \otimes y_j^\theta \in V_\theta.$$

For $z \in M_\theta \otimes U_\theta$ let $R(z)$ be the smallest subspace R of M_θ with $z \in R \otimes U_\theta$. Thus $R(w_\theta) = P_\theta$, and $R(z(a \otimes b)) = R(z)a$ whenever b is injective.

Step 5: contradiction. By Proposition 3.1(iii) there is a 3-element $s \in S$ with $ws = 2w$. By Lemma 3.7(c), $s = x^{-1}tx$ with $t \in C_S(f)$ a 3-element and $x \in Q_1$. Let $\pi \in \hat{S}$ be the image of s , which is also the image of t .

If $\pi = 1$, then $s \in N_1$, so $s \in C_0$, and $w_\theta s = 2w_\theta$ gives $y_j^\theta h_\theta(s) = 2y_j^\theta$ for all θ and j . Hence $4h(s) \in H$ fixes u_1 and u_2 , so $h(s) = 2$ and s acts on V as the scalar 2, contradicting Proposition 3.1(iv).

If $\pi \neq 1$, put $z = wx^{-1}$, so that $zt = 2z$. Here $z_\theta = w_\theta(\rho_\theta(x)^{-1} \otimes 1)$, so $R(z_\theta) = P_\theta \rho_\theta(x)^{-1}$. If t maps V_θ onto $V_{\theta'}$, then $z_\theta t = 2z_{\theta'}$, so $R(z_\theta)a_t = R(z_{\theta'})$, that is,

$$P_{\theta'} \iota_{\theta'} = P_\theta \iota_\theta \cdot (\iota_\theta^{-1} \rho_\theta(x)^{-1} \iota_\theta) \hat{a}_t (\iota_{\theta'}^{-1} \rho_{\theta'}(x) \iota_{\theta'}) \in P_\theta \iota_\theta E \mu E = P_\theta \iota_\theta Y$$

by Claim A. Hence $\chi(\theta') = \chi(\theta)$ for all θ , that is, π preserves χ , and so does the power of π of order 3. This contradicts the choice of χ . $\square$

3.6. Proof of Theorem A. A counterexample of minimal order is as in Proposition 3.1, with $p = 5$ or $p = 7$. Both cases are impossible by Propositions 3.6 and 3.10. $\square$

Corollary 3.11. *The graphs (s), (t) and (v) are not Gruenberg–Kegel graphs of finite solvable cut groups.*

Proof. They have the vertices 5 and 7 but not the edge $5 - 7$. $\square$

4. CUT GROUPS WITHOUT ELEMENTS OF ORDER 10

In this section we collect results on solvable cut groups without elements of order 10 which are needed for Theorem B. Recall (Lemma 2.8) that such a group has no element of order 105.

4.1. Modules in characteristic 5.

Lemma 4.1. *Let R be a solvable cut $\{2, 3, 7\}$ -group acting faithfully and irreducibly on an $\mathbb{F}_5$ -space M . Assume that a Sylow 2-subgroup T of R is isomorphic to Q_8 and acts without fixed points on M , and that every $m \in M$ satisfies $mg = 2m$ for some 2-element $g \in R$. Then $7 \nmid |R|$.*

Proof. The involution z of T acts as -1 , so $z \in Z(R)$ and $\langle z \rangle \leq O_2(R) \leq T$. Suppose $7 \mid |R|$.

Case $O_2(R) = T$. The elements of order 7 centralise T , as $|\text{Aut } Q_8| = 24$. By Burnside's theorem $C_R(T) = \langle z \rangle \times K$ with K of odd order, and K contains the elements of order 7. Let U be the faithful irreducible 2-dimensional $\mathbb{F}_5 T$ -module; as z acts as -1 , $M \cong U \otimes X_0$ as $T \times K$ -module, where $X_0 = \text{Hom}_T(U, M)$. Every $m \in M$ is a 2-eigenvector of a 2-element g , which lies in T and has order 4; as $\dim U_g(2) = 1$, $m = u \otimes x$ is a pure tensor. Hence $\dim X_0 = 1$, K acts on X_0 by scalars of odd order, i.e. trivially, and $K = 1$: a contradiction.

Case $O_2(R) = \langle y_1 \rangle \cong C_4$. Let $R_0 = C_R(y_1)$; then $|R : R_0| = 2$, and the Sylow 2-subgroup $\langle y_1 \rangle$ of R_0 is central in R_0 , so $R_0 = K' \times \langle y_1 \rangle$ with K' a $\{3, 7\}$ -group, and the 2-elements of R_0 are the powers of y_1 . The eigenspaces M_+ , M_- of y_1 for the eigenvalues 2, 3 are R_0 -invariant and are interchanged by any $x \in R \setminus R_0$; so $\dim M_\pm = n \geq 1$. Take $v = v_+ + v_-$ with $0 \neq v_\pm \in M_\pm$ and a 2-element y' with $vy' = 2v$. Then $y' \notin R_0$, so $y' \in xR_0$ interchanges M_+ and M_- , and $v_+y' = 2v_-$. Since v_- was arbitrary and v_+y' lies in the R_0 -orbit of v_+x , R_0 is transitive on $M_- \setminus \{0\}$, and $5^n - 1$ divides $4|K'|$. As $8 \nmid 5^n - 1$, n is odd, and if $n \geq 3$ a Zsigmondy prime [14] divisor of $5^n - 1$ has odd multiplicative order n modulo itself, so it is neither 3 nor 7 ($\text{ord}_3(5) = 2$, $\text{ord}_7(5) = 6$). So $n = 1$, $\dim M = 2$, and $|R|$ divides $|GL(2, 5)| = 480$: a contradiction.

Case $O_2(R) = \langle z \rangle$. By Lemma 2.3(2), T centralises $O_3(R)$ and $O_7(R)$ (as z does), hence $F(R) = \langle z \rangle \times O_3(R) \times O_7(R)$; so $T \leq C_R(F(R)) \leq F(R)$, a contradiction. $\square$

Lemma 4.2. *Let L be a $\{3, 7\}$ -group without elements of order 21 with $O_3(L) = 1$. Then $A = O_7(L)$ is a Sylow 7-subgroup of L , and L/A is a cyclic 3-group acting without fixed points on A .*

Proof. A nontrivial 3-element has no fixed point on A , so every 3-subgroup of L acting on A is cyclic. We have $O_7(L/A) = 1$, so $F(L/A) = O_3(L/A)$, which lifts to a 3-group acting fixed-point-freely on A ; so $F(L/A)$ is cyclic. As $C_{L/A}(F(L/A)) \leq F(L/A)$ and the automorphism group of a cyclic 3-group has order $2 \cdot 3^k$, L/A is a 3-group, hence cyclic and acting fixed-point-freely on A . $\square$

Theorem 4.3. *Let $X = P \rtimes R$ be a solvable cut group without elements of order 10, where $5 \nmid |R|$ and P is a faithful irreducible $\mathbb{F}_5 R$ -module. Then no element of order 7 of R has a nonzero fixed point on P .*

Proof. Suppose $a \in R$ has order 7 and $C_P(a) \neq 0$. By Lemma 2.1, R is a $\{2, 3, 7\}$ -group. A Sylow 2-subgroup T of R acts without fixed points on P , and $4 \mid |R|$ as elements of order 5 are rational; so $T \cong Q_8$ or $T \cong C_4$ (Lemma 2.2(3)), and its involution z acts as -1 and is central in R . By Lemma 2.2(1) and Lemma 2.7, every $v \in P$ satisfies $vg = 2v$ for a 2-element $g \in R$. So $T \not\cong Q_8$ by Lemma 4.1, and $T = \langle y \rangle \cong C_4$. By Lemma 2.2(4), X has no elements of order 12 or 28. If $O_2(R) = T$, then T is central ($R/C_R(T)$ is a 2-group and $C_R(T)$ contains a Sylow 2-subgroup), so by Lemma 2.9(b) a is rational in R , which is impossible as no 2-element inverts a . Hence $O_2(R) = \langle z \rangle$, and by Burnside's transfer theorem $R = K \rtimes \langle y \rangle$ with K a $\{3, 7\}$ -group.

R has no element of order 21. Otherwise let $r, a' \in R$ be commuting elements of orders 3 and 7. The subgroup $B_R(ra')$ of $\text{Aut}\langle r \rangle \times \text{Aut}\langle a' \rangle \cong C_2 \times C_6$ has order at least 6, and if its order is 6 it does not contain $(-1, -1)$; so it contains $(-1, 1)$ or $(1, -1)$. A 2-element inducing it is not central, hence has order 4 and centralises a' or r , giving an element of order 28 or 12.

So $O_3(K)$ and $O_7(K)$ are not both nontrivial. As a is inverse semi-rational, some 3-element $t \in K$ satisfies $a^t = a^2$.

Case $B = O_3(K) \neq 1$. Then $O_7(K) = 1$, the 7-elements act fixed-point-freely on B , so R_7 is cyclic, of order 7 (Lemma 2.2(2)). As $O_3(K/B) = 1$, $F(K/B)$ is the image of $\langle a \rangle$, and K/B embeds in $C_7 \rtimes C_3$; since $t \notin B$ (otherwise $a = a^{-1}a^t \in B$), $K/B \cong C_7 \rtimes C_3$. Thus $R/B \cong (C_7 \rtimes C_3) \rtimes C_4$ is a solvable cut group with a normal Sylow subgroup of order 7 and without elements of order 21. By [5, Lemma 3.6] its Sylow 2-subgroup is rational; but it is cyclic of order 4.

Case $A = O_7(K) \neq 1$. Then $O_3(K) = 1$, and by Lemma 4.2 A is a normal Sylow 7-subgroup of R and K/A is a cyclic 3-group. We claim:

(*) for every $b \in A$ of order 7, $N_A(\langle b \rangle)$ centralises $W := C_P(b)$.

Let $N = N_A(\langle b \rangle)$; as N is a 7-group, $N = C_A(b)$, so N preserves W . We may assume $W \neq 0$. The 3-elements of R lie in K ; let $M_b = N_K(\langle b \rangle)$, so $M_b \cap A = N$. A nontrivial 3-element of M_b does not centralise b , so $|M_b : N| \leq 3$ and $C_{M_b}(b) = N$. For $0 \neq v \in W$, $B_X(vb)$ contains $\{1\} \times \{1, 2, 4\}$, so by Lemma 2.7 there is a 3-element of M_b fixing v and inducing $b \mapsto b^2$. Hence $M_b = N\langle t_0 \rangle$ with $|M_b : N| = 3$, t_0 has order 3 (as $t_0^3 \in N$), and the elements of M_b inducing $b \mapsto b^2$ form the coset Nt_0 . As t_0 acts without fixed points on N , Nt_0 is the N -conjugacy class of t_0 . Therefore

$$W = \bigcup_{g \in N} C_W(t_0^g) = \bigcup_{g \in N} C_W(t_0)g.$$

Write $W = W_0 \oplus W'$ with $W_0 = C_W(N)$ and $W' = [W, N]$, both t_0 -invariant. The display shows that t_0 centralises W_0 and that $W' = \bigcup_{g \in N} C_{W'}(t_0)g$. Suppose $W' \neq 0$, let $m = \dim W'$ and $\bar{N} = N/C_N(W') \neq 1$. Then t_0 acts without fixed points on $\bar{N}$, hence nontrivially on W' , and the group induced by $\bar{N}\langle t_0 \rangle$ on W' is a Frobenius group with $C_{W'}(\bar{N}) = 0$. By Brauer's permutation lemma [11, Theorem 6.32], t_0 fixes no nontrivial irreducible character of $\bar{N}$, so every irreducible constituent of $W' \otimes \mathbb{F}_5$ is induced from $\bar{N}$, and $\dim C_{W'}(t_0) = m/3$. Hence $5^m - 1 \leq |\bar{N}|(5^{m/3} - 1)$, while $|\bar{N}| \leq |GL(m, 5)|_7 = 7^{\sum_{j \leq m/6} (1 + v_7(j))} < 7^{7m/36}$. As $5^{2/3} > 7^{7/36}$, this is impossible. So $W' = 0$, proving (*).

Now let $z_0 \in Z(A)$ have order 7. By (*) for $b = a$, $z_0 \in C_A(a)$ centralises $C_P(a) \neq 0$, so $C_P(z_0) \neq 0$; by (*) for $b = z_0$, $A = N_A(\langle z_0 \rangle)$ centralises $C_P(z_0)$. But $C_P(A)$ is R -invariant, so it is 0 or P ; both are impossible. $\square$

4.2. Two modules.

Proposition 4.4. *Let G be a cut group without elements of order 10, and suppose $G = (P \times W) \rtimes S$, where $P \neq 1$ is an elementary abelian normal Sylow 5-subgroup, W is a normal Sylow 7-subgroup and S is a $\{2, 3\}$ -group. Let $Z \neq 1$ be a normal elementary abelian subgroup of G contained in $Z(W)$. Then:*

- (a) *the Sylow 2-subgroups of S are isomorphic to Q_8 and centralise Z ;*
- (b) *for every $1 \neq f \in P$ there is $t_f \in C_S(f)$ of order 3 acting on Z as a scalar;*
- (c) *no nontrivial 3-element of S centralises a nontrivial element of Z .*

Proof. We write P and Z additively when convenient. By Lemma 2.7 (with $M = P \times W$), for $c \in P \cup Z \cup PZ$ and $s \in S$ commuting with c , $B_G(cs)$ is induced by $N_S(\langle c \rangle) \cap N_S(\langle s \rangle)$. Let $D = C_S(Z)$.

Step 1. As G has no element of order 10, S_2 acts without fixed points on P ; as elements of order 5 are rational, $4 \mid |S|$; so $S_2 \cong C_4$ or $S_2 \cong Q_8$ by Lemma 2.2(3). By Lemma 2.8, no nontrivial 3-element of S centralises both a nontrivial element of P and one of Z .

Step 2. Let $1 \neq f \in P$ and $1 \neq w \in Z$. Then $|fw| = 35$, and $B(fw) \leq \text{Aut}\langle f \rangle \times \text{Aut}\langle w \rangle \cong C_4 \times C_6$ has order at least 12, so it contains the squares $\{\pm 1\} \times \{1, 2, 4\}$. Hence there are a 3-element $t \in S$ with $f^t = f$ and $w^t = w^2$, and a 2-element of S inducing $(-1, 1)$. Moreover the Sylow 2-subgroup of $B(fw)$ has order at least 4, and if $|B(fw)| = 12$ it does not contain $(-1, -1)$; so it contains some $(\pm 2, \varepsilon)$, induced by an element $y \in S$ of order 4 normalising $\langle f \rangle$ and $\langle w \rangle$.

Step 3. Fix $1 \neq f \in P$. Then $P_f := C_S(f)$ is a 3-group (no elements of order 10), nontrivial by Step 2. By Step 1 its nontrivial elements have no fixed points on Z , so P_f is cyclic. For every w , the element t of Step 2 lies in P_f , and $t^3 = 1$ by Step 1. So with $\langle t_f \rangle = \Omega_1(P_f)$ we get $Z = Z_{t_f}(2) \cup Z_{t_f}(4)$, and t_f acts on Z as a scalar; we choose t_f acting as 2. This proves (b).

Step 4. The subgroup $B(ft_f)$ of $\text{Aut}\langle f \rangle \times \text{Aut}\langle t_f \rangle \cong C_4 \times C_2$ projects onto $\text{Aut}\langle f \rangle$ (a subgroup of index 2 not containing $(-1, -1)$ does). So some element of order 4, $y_f \in N_S(\langle f \rangle) \cap N_S(\langle t_f \rangle)$, induces $f \mapsto f^{\pm 2}$. Since $t_f^{y_f}$ acts on Z as the scalar 2 while t_f^{-1} acts as 4, $t_f^{y_f} = t_f$. Thus $y_f t_f$ has order 12, and Lemma 2.2(4) shows $S_2 \cong Q_8$. Its involution z acts on P as -1 .

Step 5: $z \in Z(S) \cap D$. $E_5 := C_S(P)$ is a normal 3-subgroup of S contained in every P_f , so it is cyclic. For $s \in S$, $e = z^{-1}z^s \in E_5$. If $E_5 \neq 1$, then $\Omega_1(E_5) = \langle t_f \rangle$ is normal in S and t_f acts on Z as the scalar 2, so t_f is central; then z does not invert the cyclic group E_5 , so it centralises it, and $1 = (z^s)^2 = e^2$ gives $e = 1$. Thus z is central and is the only involution of S . The 2-element inducing $(-1, 1)$ in Step 2 must be z , since an element of order 4 has square z and so induces ± 2 on $\langle f \rangle$; so z centralises every $w \in Z$.

Step 6: every element of order 4 of S acts on Z as ± 1 . We have $N_S(\langle f \rangle) = P_f \langle y_f \rangle$. As y_f centralises t_f , it does not invert the cyclic 3-group P_f , so it centralises it, and $N_S(\langle f \rangle) = P_f \times \langle y_f \rangle$, whose elements of order 4 are $y_f^{\pm 1}$. By Step 2, for each w one of them normalises $\langle w \rangle$; so y_f acts on Z as a scalar, which is ± 1 as $y_f^2 = z \in D$. Every $y \in S$ of order 4 satisfies $y^2 = z$, so it has an eigenvector on P with eigenvalue 2 or 3, and hence $y = y_f^{\pm 1}$ for some f .

Step 7: $S_2 \leq D$. By Step 6 a Sylow 2-subgroup $T \cong Q_8$ acts on Z through a homomorphism $T \rightarrow \{\pm 1\}$, whose kernel contains some $\langle y_1 \rangle \cong C_4$. Let $1 \neq w \in Z$; then $|wy_1| = 28$. By Lemma 2.3(1), every 2-element of $N_G(\langle wy_1 \rangle)$ centralises w , so $B(wy_1) \leq \text{Aut}\langle w \rangle \times \text{Aut}\langle y_1 \rangle \cong C_6 \times C_2$ contains $(1, -1)$: some 2-element x inverts y_1 and centralises w . As $x \neq z$, x has order 4, $\langle y_1, x \rangle \cong Q_8$ acts on Z by ± 1 with y_1 and x in the kernel, so $\langle y_1, x \rangle \leq D$; as $D \trianglelefteq S$, $S_2 \leq D$. This proves (a).

Step 8. Suppose a 3-element $r \in S$, which we may take of order 3, centralises some $1 \neq w \in Z$. The subgroup $B(wr)$ of $\text{Aut}\langle w \rangle \times \text{Aut}\langle r \rangle \cong C_6 \times C_2$ has even order, and it contains no element $(-1, \pm 1)$ because 2-elements centralise Z ; so it contains $(1, -1)$, and some 2-element $x \in D$ inverts r . As S/D has odd order, $r \in D$. Elements of D centralise Z , so by Step 1 D acts

without fixed points on P : D is a Frobenius complement with Sylow 2-subgroup Q_8 and cyclic Sylow 3-subgroups. If D has a normal Sylow 2-subgroup T_D , then $x \in T_D$, and $\bar{r} = \bar{r}^{-1}$ in the 3-group D/T_D , so $r \in T_D$, absurd. Otherwise $O_3(D) \neq 1$ (if $O_3(D) = 1$, then $F(D) = O_2(D)$ is $\langle z \rangle$ or cyclic of order 4, and $C_D(F(D)) \leq F(D)$ forces $|D| \leq 8$). The subgroup of order 3 of the cyclic group $O_3(D)$ is normal in D , and all subgroups of order 3 of D are conjugate; so $\langle r \rangle \trianglelefteq D$. Let $T \leq D$ be a Sylow 2-subgroup containing x . Then $T \rightarrow \text{Aut}\langle r \rangle$ is onto with kernel some $\langle y \rangle \cong C_4$, so $|yr| = 12$, while every element of $T \setminus \langle y \rangle$ normalises $\langle yr \rangle$ and inverts r . This contradicts Lemma 2.3(1), and proves (c). $\square$

4.3. Modules in characteristic 7.

Lemma 4.5. *Let X be a solvable cut group without elements of order 10, let M be an elementary abelian normal 7-subgroup of X centralised by every 7-element of X , and let $f \in X$ have order 5 with $W_f := C_M(f) \neq 1$. Put $A_f = C_X(f)$ and $H_f = N_X(\langle f \rangle)$. Then:*

- (a) *there are $t_f \in A_f$ of order 3 acting on W_f as the scalar 2, and $y_f \in H_f$ of order 4 inducing an automorphism of order 4 of $\langle f \rangle$ and centralising t_f ; in particular the Sylow 2-subgroups of X are not cyclic;*
- (b) *every involution of H_f centralises W_f .*

If $X = M \rtimes R$ and $f \in R$, then t_f and y_f can be chosen in R .

Proof. By Lemma 2.1, $\pi(X) \subseteq \{2, 3, 5, 7\}$. As X has no elements of order 10, A_f has odd order and $H_f/A_f \cong C_4$ (f is rational), so the Sylow 2-subgroups of H_f are cyclic of order 4. Let $K_f = C_{A_f}(W_f)$ and $\bar{A}_f = A_f/K_f$, which acts faithfully on W_f . By hypothesis the 7-elements of A_f lie in K_f , so $\bar{A}_f$ is a $\{3, 5\}$ -group. By Lemma 2.8, no nontrivial 3-element of A_f has a nonzero fixed point on W_f ; hence every 3-subgroup of $\bar{A}_f$ acts fixed-point-freely on W_f and is cyclic.

(a) For $1 \neq w \in W_f$, the group $B_X(fw)$, which is induced by $N_X(\langle f \rangle) \cap N_X(\langle w \rangle)$, has order at least 12 in $C_4 \times C_6$, so it contains $\{\pm 1\} \times \{1, 2, 4\}$. Hence there are a 3-element $t \in A_f$ with $w^t = w^2$, necessarily of order 3, and an involution of H_f inverting f and centralising w . By Lemma 2.11 ($r = 3$, $q = 5$), elements of order 3 of $\bar{A}_f$ centralise $O_5(\bar{A}_f)$. If $O_3(\bar{A}_f) = 1$, they would centralise $F(\bar{A}_f) = O_5(\bar{A}_f)$ and lie in it; so $O_3(\bar{A}_f) \neq 1$ is cyclic, and its subgroup $\langle \bar{t}_f \rangle$ of order 3 is the only subgroup of order 3 of $\bar{A}_f$ (any element of order 3 generates with $O_3(\bar{A}_f)$ a 3-subgroup, which is cyclic). Hence every $w \in W_f$ is an eigenvector of t_f with eigenvalue 2 or 4, and t_f acts on W_f as a scalar, which we may take to be 2. Next, $B_X(ft_f) \leq C_4 \times C_2$ projects onto $\text{Aut}\langle f \rangle$, so some $y_f \in H_f \cap N_X(\langle t_f \rangle)$ of order 4 induces $f \mapsto f^{\pm 2}$. As y_f preserves W_f , on which t_f^{-1} acts as 4, $t_f^{y_f} = t_f$. Thus $y_f t_f$ has order 12, and by Lemma 2.2(4) the Sylow 2-subgroups are not cyclic.

(b) Let $x_f = y_f^2$, an involution centralising t_f ; the involutions of H_f are the A_f -conjugates of x_f . No element of order 5 commutes with an involution, so x_f acts without fixed points on $\bar{P} := O_5(\bar{A}_f)$ (fixed points lift, by coprime action), and $\bar{P}$ is abelian and inverted by x_f . Let $C = C_{\bar{A}_f}(\bar{P})$. Then $O_5(C) = \bar{P}$, the 5-elements of C centralise the cyclic group $O_3(C)$, hence $F(C)$, so they lie in $\bar{P}$; thus $C = \bar{P} \times T_C$ with T_C a cyclic 3-group, which contains $\bar{t}_f$ and is therefore centralised by x_f . An involution of H_f has the form $x_f b$ on W_f with $b \in \bar{A}_f$ inverted by x_f ; as $x_f b$ also inverts $\bar{P}$, $b \in C$, and then $b \in \bar{P}$.

Decompose $W_f \otimes \bar{\mathbb{F}}_7$ into weight spaces for $\bar{P}$. Let λ be a nontrivial weight, of order 5^k . Since $4 \mid \text{ord}_{5^k}(7)$, the Galois class of λ contains λ^{-1} ; on the corresponding $\mathbb{F}_7$ -rational block B , the image of $\mathbb{F}_7 \bar{P}$ is the field $E = \mathbb{F}_7(\zeta_{5^k})$, and B is an E -space. Each map $x_f b$ ($b \in \bar{P}$) preserves B and is semilinear with respect to the automorphism $\zeta \mapsto \zeta^{-1}$ of order 2 of E ; being an involution, by Galois descent its fixed points form an E_0 -form of B (E_0 the fixed field), so they number $|B|^{1/2}$; and at most 5^k of these maps are distinct on B . By (a), every vector of W_f is fixed by

some involution of H_f ; hence $|B| - 1 \leq 5^k(|B|^{1/2} - 1)$, i.e. $|B|^{1/2} + 1 \leq 5^k$. But $\dim_{\mathbb{F}_7} B \geq 4 \cdot 5^{k-1}$, so $|B|^{1/2} \geq 7^{2 \cdot 5^{k-1}} > 5^k$. Therefore $\bar{P}$ acts trivially on W_f , i.e. $\bar{P} = 1$; every involution of H_f acts on W_f as x_f ; and since every vector of W_f is fixed by some involution, all involutions of H_f centralise W_f .

Finally, if $X = M \rtimes R$ and $f \in R$, then $A_f = W_f \rtimes C_R(f)$ and $H_f = W_f \rtimes N_R(\langle f \rangle)$; conjugating by an element of W_f , which does not change the action on M , we may take t_f and y_f in R . $\square$

Corollary 4.6. *Let X be a solvable cut group without elements of order 10 whose Sylow 2-subgroup T is isomorphic to Q_8 and whose Hall $\{2, 5\}$ -subgroups are Frobenius groups $P \rtimes T$ with P noncyclic. If M is an elementary abelian normal 7-subgroup of X centralised by every 7-element of X , then T and P centralise M .*

Proof. The involution z of T inverts the abelian group P , so $z \in N_X(\langle f \rangle)$ for every $1 \neq f \in P$, and z centralises $C_M(f)$ by Lemma 4.5(b). As P is noncyclic, $M = \sum_{1 \neq f \in P} C_M(f)$, so z centralises M . By Lemma 2.3(2), T centralises M , and then so does $P = [P, T]$, since $C_X(M) \trianglelefteq X$ contains all Sylow 2-subgroups. $\square$

Theorem 4.7. *Let $X = V \rtimes R$ be a solvable cut group without elements of order 10, where $7 \nmid |R|$ and V is a faithful irreducible $\mathbb{F}_7 R$ -module. Then no element of order 5 of R has a nonzero fixed point on V .*

Proof. Suppose $f \in R$ has order 5 and $W_f = C_V(f) \neq 0$. Lemma 4.5 applies with $M = V$, and we take $t_f, y_f \in R$. By Lemma 2.1, R is a $\{2, 3, 5\}$ -group. Let J be a Hall $\{2, 5\}$ -subgroup of R containing the $\{2, 5\}$ -subgroup $\langle f, y_f \rangle$. It has no element of order 10, so by the theorem of Gruenberg and Kegel [13] it is a Frobenius or a 2-Frobenius group.

Step 1: J is not a Frobenius group. Suppose it is. Its kernel is not a 2-group: otherwise f lies in a complement, a 5-group, which contains $N_J(\langle f \rangle) \ni y_f$. So $J = P_5 \rtimes S_2$ with S_2 acting fixed-point-freely; S_2 is not cyclic (Lemma 4.5(a)), so it is generalised quaternion, and $S_2 \cong Q_8$ by Lemma 2.4. Then P_5 is abelian (it admits the fixed-point-free automorphism z of order 2, which inverts it), of exponent 5 (Lemma 2.2(2)), and noncyclic (Q_8 does not act faithfully on a cyclic group). So $z \in N_R(\langle f' \rangle)$ for every $1 \neq f' \in P_5$, and by Lemma 4.5(b) (applied to every f' with $C_V(f') \neq 0$), z centralises $\sum_{f'} C_V(f') = V$: a contradiction.

Step 2: J is not a 2-Frobenius group. Suppose it is. Then its Sylow 5-subgroup is cyclic, so $R_5 = \langle f \rangle$ has order 5. Let $N = O_{5'}(R)$, a $\{2, 3\}$ -group. As $C_{R/N}(O_5(R/N)) \leq O_5(R/N)$ and f is rational, $R/N \cong C_5 \rtimes C_4$, and $C_R(f) = \langle f \rangle \times C_N(f)$, where $C_N(f)$ is a 3-group acting fixed-point-freely on W_f , hence cyclic, with subgroup of order 3 equal to $\langle t_f \rangle$.

(a) *There is no abelian normal subgroup $A \neq 1$ of R with $C_A(f) = 1$.* Otherwise $C_V(A)$ is R -invariant and proper, so 0; hence all weights of A on $V \otimes \bar{\mathbb{F}}_7$ are nontrivial and lie in $\langle f \rangle$ -orbits of length 5, and W_f is the sum of the “diagonals” D_O of these orbits O . As t_f commutes with f and acts on W_f as the scalar 2, it fixes each orbit, hence each weight, and acts as 2 on every weight space. So t_f acts on V as the scalar 2, and is central in R , contradicting Lemma 2.9(a). In particular $O_2(R) = 1$ (take $A = \Omega_1(Z(O_2(R)))$). Also $O_5(R) = 1$, as $\langle f \rangle$ is not normal in the 2-Frobenius group J . So $F(R) = O_3(R) \neq 1$.

(b) Let $A \leq O_3(R)$ be minimal normal in R , and let Ω be the set of weights of A on V . They are $\mathbb{F}_7$ -rational, so $V = \bigoplus_{\varepsilon \in \Omega} V_\varepsilon$, and Ω is a single R -orbit not containing the trivial weight. Put $\omega = 2 \in \mathbb{F}_7$ and identify weights with elements of $A^* = \text{Hom}(A, \mathbb{F}_3)$ via $\varepsilon(a) = \omega^{\langle \varepsilon, a \rangle}$. By Lemma 2.9(a), A is not central, so A^* is a nontrivial irreducible $\mathbb{F}_3 R$ -module. By (a), $C_A(f) \neq 1$, so $C_A(f) = \langle t_f \rangle$; the f -fixed weights lie in $C_{A^*}(f) = \{0, \pm \varepsilon_0\}$, which pairs perfectly with $\langle t_f \rangle$. Every weight in an f -orbit of length 5 takes the value ω at t_f , since t_f acts as 2 on W_f . If all weights took the value ω at t_f , t_f would act as a central scalar; so some $\delta \in \Omega \cap \{\pm \varepsilon_0\}$ has

$\langle \delta, t_f \rangle = -1$. Thus t_f acts as ω^2 on V_δ and as ω on all other weight spaces, and similarly for t_f^s with δs in place of δ .

If $-\delta \in \Omega$, choose s with $\delta s = -\delta$ and put $g = t_f^{-1} t_f^s \in A$; it acts as ω^2 on V_δ , as ω on $V_{-\delta}$ and trivially elsewhere, so $g^f = g$, i.e. $1 \neq g \in C_A(f) = \langle t_f \rangle$. If $\Omega = \{\pm\delta\}$, then $A = \langle t_f \rangle$, $W_f \leq V_{-\delta}$, and s inverts t_f ; then $f^s \in C_R(t_f)$, whose Sylow 5-subgroups have order 5, so $\langle f^s \rangle = \langle f \rangle^c$ with $c \in C_R(t_f)$, and $W_{f^s} = W_f c \leq V_{-\delta}$; but also $W_{f^s} = W_f s \leq V_\delta$, a contradiction. Otherwise there is an f -orbit O of length 5 in Ω , and $0 \neq D_O \leq C_{W_f}(g)$, contradicting the fact that the nontrivial elements of $\langle t_f \rangle$ have no fixed points on W_f .

If $-\delta \notin \Omega$, then $|\Omega| = 5k + 1$. As A^* is a nontrivial irreducible module, $\sum_{\varepsilon \in \Omega} \varepsilon = 0$; pairing with t_f gives $5k - 1 \equiv 0 \pmod{3}$, so $k \geq 2$. Let O_1, O_2 be two f -orbits of length 5 and choose $s_1 = 1, s_2, \dots, s_6 \in R$ with $\{\delta s_2, \dots, \delta s_6\} = O_1$. Then $h = \prod_i t_f^{s_i} \in A$ acts as ω on $V_\delta \oplus \bigoplus_{\varepsilon \in O_1} V_\varepsilon$ and trivially elsewhere; so $h^f = h$, $1 \neq h \in \langle t_f \rangle$, and $0 \neq D_{O_2} \leq C_{W_f}(h)$: the same contradiction. $\square$

4.4. Normal Sylow subgroups.

Lemma 4.8. (a) If $Y \trianglelefteq X$, then $\Gamma_{\text{GK}}(X/Y)$ is a subgraph of $\Gamma_{\text{GK}}(X)$. (b) If X is a solvable cut group with $\pi(X) = \{2, 3, 5, 7\}$ and without elements of order 10, then $\Gamma_{\text{GK}}(X)$ is (p) or (u); in particular X has elements of orders 6, 14, 15 and 35.

Proof. (a) An element of order m of X/Y lifts to an element of order divisible by m . (b) By [2, Theorem C], $\Gamma_{\text{GK}}(X)$ is one of the graphs of Figure 1; by Theorem A it contains 5 – 7, and by hypothesis it does not contain 2 – 5. $\square$

Theorem 4.9. Let X be a solvable cut group with $\pi(X) = \{2, 3, 5, 7\}$ and without elements of order 10.

- (a) If X has a normal Sylow 7-subgroup W , then every 5-element of X centralises W .
- (b) If X has a normal Sylow 5-subgroup P , then every 7-element of X centralises P .

Proof. (a) Write $X = W \rtimes R$ and $W/\Phi(W) = \bigoplus_i W_i$ with irreducible $\mathbb{F}_7 R$ -modules W_i . Fix i , let $Y_i/\Phi(W) = \bigoplus_{j \neq i} W_j$ and $K_i = C_R(W_i)$. Then $Y_i K_i \trianglelefteq X$ and $X_i := X/Y_i K_i \cong W_i \rtimes (R/K_i)$, with W_i faithful and irreducible. X_i is a solvable cut group without elements of order 10, so by Theorem 4.7 it has no element of order 35. If 5 divided $|R/K_i|$, then $\pi(X_i) = \{2, 3, 5, 7\}$ and Lemma 4.8(b) would give an element of order 35. So all 5-elements of R lie in every K_i , i.e. they centralise $W/\Phi(W)$, hence W . (b) The same, with Theorem 4.3. $\square$

Lemma 4.10. Let X be a solvable cut group without elements of order 10, with $X_2 \cong Q_8$ and X_5 abelian. Then every 7-element of X centralises $O_5(X)$.

Proof. Let $M = A/B$ be a chief factor of X with $B < A \leq O_5(X)$. As $A \leq X_5$ and X_5 is abelian, $R := X/C_X(M)$ is a $5'$ -group acting faithfully and irreducibly on M ; it is a $\{2, 3, 7\}$ -group by Lemma 2.1. A Sylow 2-subgroup acts without fixed points on M (no elements of order 10, coprime action), so the Sylow 2-subgroups of R are isomorphic to Q_8 . Every $aB \in M$ satisfies $(aB)^g = a^2 B$ for some $g \in X$, as a is rational, and so for some 2-element of R . By Lemma 4.1, $7 \nmid |R|$; so the 7-elements act trivially on every such chief factor, hence on $O_5(X)$. $\square$

5. PROOF OF THEOREM B

Throughout this section, G denotes a solvable cut group of minimal order with $\Gamma_{\text{GK}}(G) = (\text{u})$. Thus G has elements of orders 21 and 35 and none of order 10, and by Lemma 2.8 none of order 105. Theorem B asserts that no such G exists. Indeed, if X is a solvable cut group with elements of orders 21 and 35 and none of order 10, then $\pi(X) = \{2, 3, 5, 7\}$ and $\Gamma_{\text{GK}}(X) = (\text{u})$ by Lemma 4.8(b).

5.1. First reductions.

Lemma 5.1. *Let $1 \neq Y \trianglelefteq G$ with $5 \in \pi(G/Y)$. Then G/Y has no element of order 21. Consequently:*

- (a) $O_2(G) = 1$, and every minimal normal 5-subgroup of G is a Sylow subgroup;
- (b) if N is a minimal normal 3-subgroup of G , then every 3-element commuting with a nontrivial 7-element lies in N ; in particular N is the only minimal normal 3-subgroup;
- (c) if N is a minimal normal 7-subgroup of G which is not a Sylow subgroup, then every 7-element commuting with a nontrivial 3-element lies in N , and N is the only such subgroup.

Proof. If G/Y had an element of order 21, then $\pi(G/Y) = \{2, 3, 5, 7\}$ by Lemma 2.1, G/Y would have no element of order 10, and Lemma 4.8(b) would give $\Gamma_{\text{GK}}(G/Y) = (u)$, against minimality. (a) If N is a minimal normal 2-subgroup, or a minimal normal 5-subgroup which is not Sylow, then $5 \in \pi(G/N)$ and images of elements of order 21 have order 21. (b) If r commutes with $w \neq 1$ of order a power of 7, the images of r and w in G/N commute and wN has order $|w|$, so $rN = 1$. A second minimal normal 3-subgroup N' would also contain the 3-parts of all elements of order 21, but $N \cap N' = 1$. (c) is proved in the same way. $\square$

Theorem 5.2. *Neither the Sylow 5-subgroup nor the Sylow 7-subgroup of G is normal. Consequently $O_2(G) = O_5(G) = 1$, $F(G) = O_3(G) \times O_7(G)$ with $O_7(G)$ not a Sylow subgroup, and $G/F(G)$ is a solvable cut group with graph (p).*

Proof. Step 1: the Sylow 7-subgroup W is not normal. Suppose it is, and write $G = W \rtimes R$. By Theorem 4.9(a), the 5-elements centralise W . Since $Z(W)$ is a central Sylow subgroup of $C_G(W)$, we have $C_G(W) = Z(W) \times L$ with $L = O_{7'}(C_G(W)) \trianglelefteq G$, and L contains all 5-elements. So $F(L)$ is a nontrivial nilpotent normal 7'-subgroup of G , and as $O_2(G) = 1$ (Lemma 5.1(a)), $O_3(G) \neq 1$ or $O_5(G) \neq 1$.

Case $O_5(G) \neq 1$. By Lemma 5.1(a), $P := O_5(G)$ is an elementary abelian Sylow 5-subgroup, which centralises W , and $G = (P \times W) \rtimes S$ with S a Hall $\{2, 3\}$ -subgroup. If W is minimal normal, put $Z = W$. Otherwise let $N_7 < W$ be minimal normal; then $N_7 \leq Z(W)$ is not Sylow, and we put $Z = \Omega_1(Z(W)) \geq N_7$. In both cases, the 7-part of every element of order 21 lies in Z (by Lemma 5.1(c) in the second case). Conjugating an element of order 21 so that its 3-part lies in S , we obtain a nontrivial 3-element of S centralising a nontrivial element of Z . This contradicts Proposition 4.4(c).

Case $O_5(G) = 1$. Then $O_3(G) \neq 1$; let $N \leq O_3(G)$ be minimal normal. As $O_3(G)$ commutes with W , Lemma 5.1(b) gives $O_3(G) = N$. Moreover W is minimal normal: otherwise a minimal normal $N_7 < W$ is not Sylow, and Lemma 5.1(c) gives $W \leq N_7$, as W commutes with N . So $V := W$ is elementary abelian and $F(G) = N \times V$. Each element f of order 5 centralises V , so it acts without fixed points on N (no elements of order 105); hence G_5 is cyclic, of order 5. Take $f \in R$. By Lemma 4.5(a) (with $X = G = V \rtimes R$, $M = V$ and $C_V(f) = V$), the Sylow 2-subgroups of G are not cyclic, and there is $y_f \in R$ of order 4 normalising $\langle f \rangle$. A Hall $\{2, 5\}$ -subgroup J of R containing $\langle f, y_f \rangle$ has no element of order 10, so it is a Frobenius or 2-Frobenius group. It is not Frobenius: a kernel which is a 2-group is excluded as in Step 1 of the proof of Theorem 4.7, and a noncyclic 2-group cannot act faithfully on C_5 . So J is 2-Frobenius; as $|J|_5 = 5$, its lower Frobenius kernel $F_1 = O_2(J)$ is nontrivial and f acts on it without fixed points. Let $U = \Omega_1(Z(F_1))$. Since f lies in the normal subgroup $C_R(V)$ of R , $U = [U, f] \leq C_R(V)$. As $C_G(N \times V) \leq N \times V$, U acts faithfully on N , and $U\langle f \rangle$ is a Frobenius group. By Lemma 2.5, f fixes some $1 \neq n \in N$, and for $1 \neq v \in V$ the elements n, f, v commute pairwise: an element of order 105.

Step 2: the Sylow 5-subgroup P is not normal. Suppose it is. By Lemma 5.1(a), $P = O_5(G)$ is elementary abelian, and by Theorem 4.9(b) every 7-element centralises P . If $O_3(G) \neq 1$, the 3-part r of an element rw of order 21 lies in a minimal normal 3-subgroup (Lemma 5.1(b)), so

r , w and any $1 \neq f \in P$ commute pairwise: an element of order 105. So $O_3(G) = 1$. Let R be a Hall $5'$ -subgroup and $L = C_R(P) = O_{5'}(C_G(P)) \trianglelefteq G$. Then L contains all 7-elements, has odd order (no elements of order 10), has no element of order 21 (no elements of order 105), and $O_3(L) \leq O_3(G) = 1$. By Lemma 4.2, $O_7(L)$ is a Sylow 7-subgroup of G , and it is normal in G ; this contradicts Step 1.

Step 3. $O_2(G) = 1$ by Lemma 5.1(a); if $O_5(G) \neq 1$, it would contain a minimal normal 5-subgroup, which would be a normal Sylow subgroup. So $F(G) = O_3(G) \times O_7(G)$, and $O_7(G)$ is not Sylow by Step 1. Thus $5, 7 \in \pi(G/F(G))$, so $\pi(G/F(G)) = \{2, 3, 5, 7\}$, and by Lemma 4.8(b) and the minimality of G , $\Gamma_{\text{GK}}(G/F(G)) = (p)$. $\square$

5.2. Solvable cut groups with graph (p).

Lemma 5.3. *Let H be a solvable cut group with $\Gamma_{\text{GK}}(H) = (p)$. Then H has a quotient $H_0 = (N_5 \times C_7) \rtimes S_0$, where $S_0 = Q_8 \rtimes C_3$ (that is, $Q_8 \times C_3$ or $SL(2, 3)$), N_5 is a noncyclic elementary abelian 5-group on which $Q_8 = O_2(S_0)$ acts without fixed points, Q_8 centralises C_7 , and C_3 acts on C_7 by $a \mapsto a^2$. Moreover $F(H_0) = N_5 \times C_7$, and N_5 is a minimal normal subgroup of H_0 , i.e. an irreducible $\mathbb{F}_5 S_0$ -module. In particular $|H_0|_2 = 8$, $|H_0|_3 = 3$ and $|H_0|_7 = 7$.*

Proof. Choose $Y \trianglelefteq H$ maximal with $\Gamma_{\text{GK}}(H/Y) = (p)$ and put $H_0 = H/Y$. Then no nontrivial normal subgroup N of H_0 satisfies $\Gamma_{\text{GK}}(H_0/N) = (p)$, so by [5, Lemma 3.3] every nontrivial normal p -subgroup of H_0 is an elementary abelian Sylow subgroup with $p \in \{5, 7\}$. Hence $F(H_0)$ is N_5 , N_7 or $N_5 \times N_7$, where N_p is a normal Sylow p -subgroup, minimal normal in H_0 , and $H_0 = F(H_0) \rtimes S$ acts faithfully on $F(H_0)$. If $F(H_0) = N_7$, then Theorem 4.7 contradicts $5-7 \in \Gamma_{\text{GK}}(H_0)$; if $F(H_0) = N_5$, Theorem 4.3 does. So $H_0 = (N_5 \times N_7) \rtimes S$ with S a $\{2, 3\}$ -group, and Proposition 4.4 (with $P = N_5$ and $W = Z = N_7$) shows that the Sylow 2-subgroups of S are isomorphic to Q_8 and lie in $D = C_S(N_7)$, and that some element t_f of order 3 acts on N_7 as a scalar. Since $3-7 \notin \Gamma_{\text{GK}}(H_0)$, the 3-elements of S have no fixed points on N_7 ; so D is a 2-group, $D = S_2 \cong Q_8$ is normal in S , and S/D is a 3-group acting faithfully and without fixed points on N_7 , hence cyclic. As S/D is a quotient of H_0 , it is cut, so $|S/D| = 3$, and S/D is generated by the image of t_f . Thus H_0 acts on the minimal normal subgroup N_7 by scalars, and $N_7 \cong C_7$. Finally N_5 is not cyclic, as Q_8 acts on it faithfully, and it is minimal normal in H_0 by [5, Lemma 3.3]. $\square$

Corollary 5.4. *$G/F(G)$ has a quotient H_0 as in Lemma 5.3 with kernel of odd order. The Sylow 5-subgroup P of G is elementary abelian and noncyclic, the Hall $\{2, 5\}$ -subgroups of G are Frobenius groups $P \rtimes T$ with $T \cong Q_8$, and the involution z of T inverts P .*

Proof. Apply Lemma 5.3 to $G/F(G)$. As $F(G)$ is a $\{3, 7\}$ -group, P maps onto the noncyclic group N_5 . A Hall $\{2, 5\}$ -subgroup J of G has no element of order 10, so it is a Frobenius or a 2-Frobenius group. It is not 2-Frobenius, as its Sylow 5-subgroups are not cyclic. If it is Frobenius, its kernel is not a 2-group: we may choose J to contain an element f of order 5 and a 2-element normalising $\langle f \rangle$ and inducing an automorphism of order 4 (f is rational), and this 2-element would lie in the complement containing f , a 5-group. So $J = P \rtimes T$ is a Frobenius group with kernel P and complement a Sylow 2-subgroup T , which maps onto Q_8 ; hence T is generalised quaternion, and $T \cong Q_8$ by Lemma 2.4. Thus the kernel of $G \rightarrow H_0$ has odd order, z inverts P , and P is abelian, of exponent 5 by Lemma 2.2(2). $\square$

From now on, P and $T \cong Q_8$ are as in Corollary 5.4, and $\bar{G} = G/F(G)$; bars denote images in $\bar{G}$.

5.3. The structure of $F(G)$.

Lemma 5.5. *If $O_3(G) \neq 1 \neq O_7(G)$, then $F(G) = N \times V$ with $N = O_3(G)$ and $V = O_7(G)$ minimal normal in G , and $|\bar{G}|_3 = 3$, $|\bar{G}|_7 = 7$.*

Proof. Let $N \leq O_3(G)$ and $V \leq O_7(G)$ be minimal normal; V is not Sylow by Theorem 5.2. As $O_3(G)$ commutes with V and $O_7(G)$ commutes with N , Lemma 5.1(b),(c) give $O_3(G) = N$ and $O_7(G) = V$. By the same lemma, 3-elements outside N act without fixed points on V , and 7-elements outside V act without fixed points on N . So the Sylow 3- and 7-subgroups of $\bar{G}$ act fixed-point-freely on V and N , respectively; hence they are cyclic, and by Lemma 2.2(2) of prime order. $\square$

Proposition 5.6. *$O_3(G)$ and $O_7(G)$ are not both nontrivial.*

Proof. Otherwise, by Lemma 5.5, $F(G) = N \times V$ and $|\bar{G}|_3 = 3$, $|\bar{G}|_7 = 7$. By Corollary 5.4, $\bar{G}$ has a quotient H_0 whose kernel $\bar{Y}$ has odd order; comparing orders, $\bar{Y}$ is a 5-group. By Lemma 4.10 (applied to $\bar{G}$, whose Sylow 2-subgroup is $\bar{T} \cong Q_8$ and whose Sylow 5-subgroup is abelian), a Sylow 7-subgroup $\langle \bar{x} \rangle$ of $\bar{G}$ centralises $\bar{Y} \leq O_5(\bar{G})$. So the preimage $\bar{Y} \times \langle \bar{x} \rangle$ of the normal subgroup C_7 of H_0 has $\langle \bar{x} \rangle$ as a characteristic subgroup, and $\langle \bar{x} \rangle = O_7(\bar{G})$. As $N \times V \leq C_G(V)$, $G/C_G(V)$ is a quotient of $\bar{G}$ acting faithfully and irreducibly on V , so its O_7 is trivial, and $\bar{x}$ acts trivially on V : every 7-element of G centralises V . By Corollary 4.6 (with $X = G$ and $M = V$), P centralises V . Then every $1 \neq f \in P$ acts without fixed points on N (a fixed point n would give pairwise commuting n, f, v with $1 \neq v \in V$), so P is cyclic, contradicting Corollary 5.4. $\square$

Proposition 5.7. *Either (B) $F(G) = Q_3 := O_3(G)$, or (C) $F(G) = W := O_7(G)$, which is not a Sylow subgroup. In either case G has a unique minimal normal subgroup N , $N \leq Z(F(G))$, and $O_2(\bar{G}) = 1$. Moreover:*

- (i) *in case (C), $O_3(\bar{G}) \neq 1$, the Sylow 7-subgroups of $\bar{G}$ have order 7 and act without fixed points on $O_3(\bar{G})$, and P does not centralise N ;*
- (ii) *in case (B), $O_7(\bar{G}) \neq 1$, $\bar{G}$ has no normal Sylow 7-subgroup, and P does not centralise N ;*
- (iii) *in both cases, $C_T(N) = 1$.*

Proof. The first statement is Theorem 5.2 and Proposition 5.6. Every minimal normal subgroup lies in $F(G)$, and Lemma 5.1(b),(c) allows only one; it lies in $Z(F(G))$. The group $O_2(\bar{G})$ meets the kernel of $\bar{G} \rightarrow H_0$ trivially and maps into $O_2(H_0)$, which is trivial as it centralises N_5 , on which Q_8 acts faithfully. By Lemma 4.10, the 7-elements of $\bar{G}$ centralise $O_5(\bar{G})$. In case (C), $O_7(\bar{G}) = 1$, so $F(\bar{G}) = O_3(\bar{G}) \times O_5(\bar{G})$ and the 7-elements, which do not lie in $F(\bar{G})$, act nontrivially on $O_3(\bar{G})$, and fixed-point-freely as $\bar{G}$ has no element of order 21; so the Sylow 7-subgroups of $\bar{G}$ are cyclic, of order 7. In case (B), $O_3(\bar{G}) = 1$; if also $O_7(\bar{G}) = 1$, then $F(\bar{G}) = O_5(\bar{G})$ would be centralised by the 7-elements, which would then lie in $F(\bar{G})$, absurd; so $O_7(\bar{G}) \neq 1$.

Let $K = C_G(N) \trianglelefteq G$. Then $N \leq Z(K)$, so K has no element of order 15 in case (C), and none of order 35 in case (B), since its product with a nontrivial element of N would have order 105.

(i) Suppose $P \leq K$. Both $\bar{B} := O_3(\bar{G})$ and K/W are normal in $\bar{G}$. If some $1 \neq \bar{f} \in \bar{P}$ centralised some $1 \neq \bar{b} \in \bar{B} \cap K/W$, then by Schur–Zassenhaus the preimage in K of $\langle \bar{b} \rangle \times \langle \bar{f} \rangle$ would contain an element of order 15. So the noncyclic group $\bar{P}$ acts without fixed points on $\bar{B} \cap K/W$, which is therefore trivial; hence $[\bar{B}, K/W] = 1$ and $\bar{B}$ centralises $\bar{P}$. Fix $1 \neq f \in P$. By coprime action there is a 3-subgroup $B_f \leq C_G(f)$ mapping onto $\bar{B}$; as f centralises N , the nontrivial elements of B_f have no fixed points on N (no elements of order 105), so $\bar{B}$ is cyclic and has no automorphism of order 7. This contradicts the fixed-point-free action of the Sylow 7-subgroups on $\bar{B}$.

(ii) We first show that $\bar{G}$ has no normal Sylow 7-subgroup. If it had one, then by Theorem 4.9(a) every 5-element of $\bar{G}$ would centralise it, and by coprime action (over the 3-group Q_3), for every 7-element w of G , $C_G(w)$ would contain a noncyclic 5-group. This group acts on $C_N(w)$, so $C_N(w) = 1$ (no elements of order 105) for all w ; but by Lemma 5.1(b) every element of order 21 of G has the form rw with $1 \neq r \in N \cap C_G(w)$. Now suppose $P \leq K$ and let $\bar{A} = O_7(\bar{G})$. As

in (i), $\bar{P}$ acts without fixed points on $\bar{A} \cap K/Q_3$, so $[\bar{A}, K/Q_3] = 1$, and the lifts $A_f \leq C_G(f)$ of $\bar{A}$ act without fixed points on N ; so $\bar{A}$ is cyclic, say of order 7^k . Then $B_{\bar{G}}(\bar{a}) = \bar{G}/C_{\bar{G}}(\bar{A})$ for a generator $\bar{a}$ of $\bar{A}$; it is a cyclic cut group, hence of order at most 6, whereas the cut condition requires $|B_{\bar{G}}(\bar{a})| \geq 3 \cdot 7^{k-1}$. So $|\bar{A}| = 7$. A 7-element of $\bar{G}$ centralises $\bar{A}$ and, by Lemma 4.10, $O_5(\bar{G})$, hence $F(\bar{G})$, so it lies in $\bar{A}$; thus $\bar{A}$ is a normal Sylow 7-subgroup of $\bar{G}$, which we have excluded.

(iii) If T centralised N , so would $P = [P, T]$, contrary to (i) and (ii). By Lemma 2.3(2), $C_T(N) = 1$. $\square$

5.4. The two final cases.

Proposition 5.8. *Case (C) of Proposition 5.7 does not occur.*

Proof. Let N be the minimal normal subgroup, a 7-group, $K = C_G(N) \geq W$ and $\hat{G} = G/K$. If every 7-element of G centralised N , then by Corollary 4.6 (with $X = G$, $M = N$) P would centralise N , contrary to Proposition 5.7(i). As the Sylow 7-subgroups $\langle \bar{x} \rangle$ of $\bar{G}$ are conjugate and $K/W \trianglelefteq \bar{G}$, we get $\bar{x} \notin K/W$, i.e. $\hat{x} \neq 1$.

Step 1: N is a free $\mathbb{F}_7\langle \hat{x} \rangle$ -module. As $\hat{x} \neq 1$, $P \not\leq K$ (Proposition 5.7(i)) and $\hat{G}$ is a quotient of $\bar{G}$, we have $5, 7 \in \pi(\hat{G})$ and $\Gamma_{\text{GK}}(\hat{G}) = (p)$ by Lemma 4.8. By Proposition 5.7(iii), $T \cap K = 1$, so the Sylow 2-subgroups of $\hat{G}$ are isomorphic to Q_8 ; $O_2(\hat{G})$ is a section of Q_8 , so it is centralised by the 7-elements; and $O_7(\hat{G}) = 1$, as N is a faithful irreducible $\mathbb{F}_7\hat{G}$ -module. By Lemma 4.10, the 7-elements of $\hat{G}$ centralise $O_5(\hat{G})$; as $C_{\hat{G}}(F(\hat{G})) \leq F(\hat{G})$, $\hat{x}$ acts nontrivially, hence without fixed points (no elements of order 21), on $\hat{B} := O_3(\hat{G}) \neq 1$. Now $C_N(\hat{B}) = 0$, as it is $\hat{G}$ -invariant, and by Brauer's permutation lemma $\hat{x}$ fixes no nontrivial irreducible character of $\hat{B}$. The Galois orbits of irreducible characters of the 3-group $\hat{B}$ over $\mathbb{F}_7$ have 3-power length, so $\hat{x}$ fixes no homogeneous component of $N|_{\hat{B}}$. Hence these components are permuted in orbits of length 7, and N is free over $\langle \hat{x} \rangle$.

Step 2: a 3-element centralising a noncyclic 5-group. By Corollary 5.4, $\bar{G} \rightarrow H_0$ has kernel $\bar{Y}$ of odd order, and $|\bar{G}|_7 = 7 = |H_0|_7$, so $\bar{Y}$ is a 7'-group. Let c be the image of $\bar{x}$, a generator of the C_7 of H_0 . Then $C_{\bar{G}}(\bar{x})$ maps onto $C_{H_0}(c) \geq N_5$: if $\bar{g}$ maps into $C_{H_0}(c)$, then $\bar{x}$ and $\bar{x}^{\bar{g}}$ are elements of order 7 of $\bar{Y}\langle \bar{x} \rangle$ with the same image, so $\bar{x}^{\bar{g}} = \bar{x}^y$ for some $y \in \bar{Y}$. Let $\bar{P}_1$ be a Sylow 5-subgroup of $C_{\bar{G}}(\bar{x})$; it maps onto N_5 . As $C_{\bar{G}}(\bar{x})$ has no 3-elements and $B_{\bar{G}}(\bar{x}) \supseteq \{1, 2, 4\}$, the Frattini argument gives a 3-element $\bar{t} \in N_{\bar{G}}(\langle \bar{x} \rangle) \cap N_{\bar{G}}(\bar{P}_1)$ with $\bar{x}^{\bar{t}} = \bar{x}^2$. Its image in H_0 has order 3, so it is conjugate to a generator of the subgroup C_3 of S_0 . The fixed points of the latter on N_5 form a group $C_5 \times C_5$: indeed N_5 is an irreducible $\mathbb{F}_5 S_0$ -module on which Q_8 acts faithfully and on which C_3 has fixed points (as $3 - 5 \in \Gamma_{\text{GK}}(H_0)$), so either $S_0 = Q_8 \times C_3$ and N_5 is the natural 2-dimensional module of Q_8 with C_3 acting trivially, or $S_0 = SL(2, 3)$ and N_5 is the 4-dimensional irreducible module $U \otimes X_2$ (U natural, X_2 the 2-dimensional irreducible $\mathbb{F}_5 C_3$ -module) on which C_3 has a 2-dimensional fixed space. By coprime action $C_{\bar{P}_1}(\bar{t})$ maps onto this group, so it is not cyclic. Applying Schur–Zassenhaus to the preimage in G of the $\{3, 5\}$ -group $\langle \bar{t} \rangle \times C_{\bar{P}_1}(\bar{t})$, an extension of the 7-group W , we obtain a 3-element $t \in G$ mapping to $\bar{t}$ and a noncyclic 5-subgroup $P_t \leq C_G(t)$.

Step 3: contradiction. Since $\hat{x}^{\hat{t}} = \hat{x}^2$ and $\hat{t}^3$ centralises $\hat{x} \neq 1$, while $\hat{G}$ has no element of order 21, the image $\hat{t}$ has order 3 and $\langle \hat{x}, \hat{t} \rangle \cong C_7 \rtimes C_3$. By Step 1, N is a projective $\mathbb{F}_7[C_7 \rtimes C_3]$ -module, i.e. a direct sum of modules $\text{Ind}_{C_3}^{C_7 \rtimes C_3}(\chi)$; by Mackey's formula the restriction of each of them to C_3 is $\chi \oplus 2\mathbb{F}_7 C_3$, which contains the trivial module. So $C_N(t) \neq 0$. The noncyclic group P_t acts on $C_N(t)$, so some $1 \neq f \in P_t$ fixes some $1 \neq w \in C_N(t)$. Then t, f, w commute pairwise and tfw has order 105, which is impossible. $\square$

Proposition 5.9. *Case (B) of Proposition 5.7 does not occur.*

Proof. Here $\bar{G} = G/Q_3$ has $O_2(\bar{G}) = O_3(\bar{G}) = 1$, so $F(\bar{G}) = O_5(\bar{G}) \times \bar{A}$ with $\bar{A} = O_7(\bar{G})$.

Step 1: every chief factor $M = A_j/A_{j+1}$ of $\bar{G}$ with $A_j \leq \bar{A}$ is centralised by all 7-elements of $\bar{G}$. Let $R_M = \bar{G}/C_{\bar{G}}(M)$ and suppose it contains an element $\hat{x}$ of order 7. Then $O_7(R_M) = 1$, and $\hat{x}$ acts nontrivially on $F(R_M)$, and trivially on $O_2(R_M)$, a section of Q_8 . If $\hat{x}$ acts nontrivially on $O_3(R_M)$, it does so without fixed points (R_M has no element of order 21), and as in Step 1 of the proof of Proposition 5.8, M is free over $\langle \hat{x} \rangle$; since R_M is cut there is a 3-element $\hat{t}$ with $\hat{x}^{\hat{t}} = \hat{x}^2$, which has order 3 as R_M has no element of order 21, and as in Step 3 there, $C_M(\hat{t}) \neq 0$. Lifting $\hat{t}$ to a 3-element of $\bar{G}$ and using coprime action, we obtain a 3-element of $\bar{G}$ centralising a nontrivial element of A_j , i.e. an element of order 21 of $\bar{G}$, which is impossible. So $\hat{x}$ acts nontrivially on $O_5(R_M)$. If $\bar{T}$ acts faithfully on M , this contradicts Lemma 4.10 applied to R_M . Otherwise Lemma 2.3(2), applied to $\bar{G}/A_{j+1}$, gives $\bar{T} \leq C_{\bar{G}}(M)$, hence $\bar{P} = [\bar{P}, \bar{T}] \leq C_{\bar{G}}(M)$ and $O_5(R_M) = 1$: again a contradiction.

Step 2: $\bar{P}$ centralises $\bar{A}$. By Step 1 and Corollary 4.6 (applied to $X = \bar{G}/A_{j+1}$ and $M = A_j/A_{j+1}$), $\bar{P}$ centralises every chief factor of $\bar{G}$ below $\bar{A}$, hence $\bar{A}$.

Step 3: $\bar{P} \trianglelefteq \bar{G}$. Let $\bar{D} = C_{\bar{G}}(\bar{A}) \trianglelefteq \bar{G}$; it contains $\bar{P}$ and $O_5(\bar{G}) \leq \bar{P}$. Now $F(\bar{D}) \leq F(\bar{G}) \cap \bar{D} = O_5(\bar{G}) \times Z(\bar{A})$, and the abelian group $\bar{P}$ centralises both factors. So $\bar{P} \leq C_{\bar{D}}(F(\bar{D})) \leq F(\bar{D})$, whence $\bar{P} = O_5(\bar{G})$.

Step 4: contradiction. By Theorem 4.9(b), every 7-element of $\bar{G}$ centralises $\bar{P}$. So, by coprime action over Q_3 , for every 7-element w of G the group $C_G(w)$ contains a noncyclic 5-group, which acts on $C_N(w)$; as there are no elements of order 105, $C_N(w) = 1$. But by Lemma 5.1(b) every element of order 21 of G has the form rw with $1 \neq r \in N \cap C_G(w)$. $\square$

5.5. Proof of Theorem B and of Corollary C. If Theorem B failed, there would be a group G as at the beginning of this section. By Theorem 5.2 and Proposition 5.6, G is in case (B) or (C) of Proposition 5.7, which are impossible by Propositions 5.9 and 5.8. This proves Theorem B, and shows that (u) is not the Gruenberg–Kegel graph of a solvable cut group. Together with Corollary 3.11 and [2, Theorems A and C], this proves Corollary C. $\square$

APPENDIX A. COMPUTER SEARCHES

The proofs above do not use computer calculations. The following computations, done with GAP 4.12 [6] and its SmallGroups and TransitiveGroups libraries, guided the work and are consistent with the results.

Cut tests. Cut-ness of a group was tested on conjugacy classes: g is inverse semi-rational if and only if g^k is conjugate to g or g^{-1} for every k prime to $|g|$. For $X = V \rtimes S$ with V an $\mathbb{F}_p S$ -module and $p \nmid |S|$, cut-ness was tested through Lemma 2.7: for each class representative $s \in S$ and each $N_S(\langle s \rangle)$ -orbit of lines in $C_V(s)$, we computed the image of the stabiliser in $\mathbb{F}_p^\times \times (\mathbb{Z}/|s|\mathbb{Z})^\times$. This routine was validated on groups realising (p) and on non-cut examples. Before the full test, modules were filtered by the necessary condition $\sum_s (p^{\dim V_s(2)} - 1) \geq |V| - 1$ coming from Proposition 3.1(iii).

Searches for Theorem A.

- (1) $p = 7$: all solvable cut groups S of order $2^a 3^b 5 \leq 1440$ with complete graph on $\{2, 3, 5\}$, noncyclic Sylow 2-subgroup and elements of order 4 (79 groups), and all irreducible $\mathbb{F}_7 S$ -modules on which elements of order 5 act without fixed points and some 2- and 3-elements have fixed points (904 modules, of dimensions 4 to 32). In every case the counting condition already fails. We also examined the Borel subgroup $2^{2+4}:(15:4)$ of $PTU(3, 4)$, a solvable cut group with complete graph on $\{2, 3, 5\}$ whose Sylow 2-subgroup has the structure described by Lemma 3.8 (the nontrivial constituents have quotients 2_-^{1+4}); its $\mathbb{F}_7$ -modules fail property (iii) of Proposition 3.1.

- (2) $p = 5$: all such S of order $2^a 3^b 7 \leq 1512$ with complete graph on $\{2, 3, 7\}$ (19 groups), with the same outcome for the 165 relevant $\mathbb{F}_5$ -modules (dimensions 6 to 48); and no subgroup of the Hall $\{2, 3, 7\}$ -subgroup of $\Gamma L(1, 5^6)$ gives a cut group $\mathbb{F}_{5^6} \rtimes S$ of the required kind.

Checks of auxiliary statements.

- (3) In the notation of the proof of Proposition 3.10: $|N_{GL(4,7)}(E)| = 11520$, $|C_N(f_0)| = 30$, $|Y| = 480$, and Y has exactly 6 orbits on the 400 points and 46 orbits on the 2850 lines of $\mathbb{F}_7^4$ (Claim B).
- (4) Lemma 3.8 was tested on all groups $X \rtimes C_5$ of orders 80, 160 and 320 with X a normal Sylow 2-subgroup and $[X, C_5] \neq 1$, and on $(E \times E) \rtimes C_5$, $(E \circ E) \rtimes C_5$ (several actions) and the Sylow 2-subgroup of $SU(3, 4)$ extended by C_5 .
- (5) Lemma 3.9: a random search found a good 6-colouring (in fact a good 3-colouring) for each of the 5456 transitive solvable $\{2, 3\}$ -groups of degree 2, 3, 4, 6, 8, 9, 12, 16, 18 and 27. With 2 colours the statement fails, for instance for $S_4 \wr S_3$ of degree 12.

Searches for Theorem B.

- (6) All cut $\{2, 3\}$ -groups S with Sylow 2-subgroup C_4 or Q_8 and elements of order 6, $|S| \leq 324$, together with all irreducible $\mathbb{F}_5 S$ -modules of dimension at most 6 and $\mathbb{F}_7 S$ -modules of dimension at most 4 satisfying the obvious edge conditions: no cut group $(N_5 \times N_7) \rtimes S$ with graph (u) arises (compare Proposition 4.4).
- (7) All solvable cut groups with graph $\{2 - 3, 3 - 5\}$ of order $2^a 3^b 5^c \leq 1800$, and those of order 960: no $\mathbb{F}_7$ -module gives a cut group with graph (u) (compare Theorem 4.7).
- (8) The only solvable cut group of order $8 \cdot 3^b 7^c \leq 1764$ with complete graph on $\{2, 3, 7\}$ and Sylow 2-subgroup Q_8 or C_4 is $(C_3^2 \rtimes Q_8) \times (C_7 \rtimes C_3)$, whose involution is not central; so it does not act faithfully and fixed-point-freely on any $\mathbb{F}_5$ -module (compare Theorem 4.3).

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